

Verification and validation of fluid flows

Numerical analysis of internal flows
and experimental characterisation of a circular-cylinder wake

— Antonio Arbizzani Tonelli

Project at a glance

Numerical and experimental study of two canonical flows, organised as a verification and validation chain. Every reported quantity is tied to a reference solution or measurement, and an explicit error metric.

LAMINAR PIPE ·
VERIFICATION

1.24%

Pressure-gradient error against
Hagen–Poiseuille; profile error
3.18%

TURBULENT PIPE ·
VALIDATION

0.082%

Friction-factor error of standard
 $k-\varepsilon$ at $Re_b = 75,000$

CYLINDER WAKE · SHEDDING

***St* 0.18–0.20**

Identified independently from
PSV velocity and lift force

CYLINDER FORCES ·
UNCERTAINTY

1.50 ± 0.08

Mean drag coefficient with first-
order standard uncertainty

METHODS

RANS CFD (PHOENICS) · grid-sensitivity analysis · particle-shadow velocimetry · spectral analysis · load-cell calibration · uncertainty propagation · MATLAB data processing

Research objective

Establish a traceable verification and validation chain for numerical pipe flow and experimental cylinder-wake measurements.

CASE 1 · INTERNAL FLOW

Pipe flow, CFD

Regimes	Laminar, $Re_b = 193$ · turbulent, $Re_b = 75,000$
Reference	Hagen–Poiseuille solution; measured profiles and friction factor
Test	Verification of the solver, validation of four $k-\epsilon$ closures
Compared	Velocity profile, pressure gradient, Darcy friction factor

CASE 2 · SEPARATED FLOW

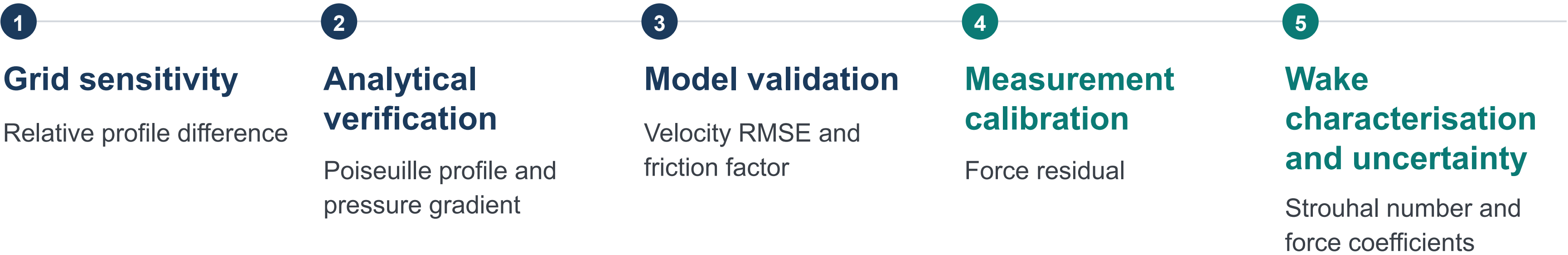
Cylinder wake, experiment

Regime	Unsteady wake, $Re_D \approx 104$
Reference	No closed-form solution: calibrated measurement chain
Test	Characterisation of the wake with quantified uncertainty
Compared	Mean field, Strouhal number, force coefficients

Verification and validation framework

NUMERICAL · PHOENICS PIPE FLOW

EXPERIMENTAL · FLUME, CIRCULAR CYLINDER



Verification — does the solver reproduce a known solution?

Validation — does the model reproduce measurements?

Grid sensitivity — necessary, but not proof of physical accuracy.

Internal flow CFD

Grid sensitivity, analytical verification of laminar flow, and validation of four k – ε closures against turbulent pipe measurements.

06 Numerical setup 07 Laminar grid 08 Turbulent grid 09 Laminar development 10 Poiseuille profile
11 Pressure gradient 12 Turbulent development 13 Turbulent structure 14 Wall scaling
15 Profile validation 16 Friction factor

Numerical setup

Parameter	Laminar	Turbulent
Diameter D	0.15 m	0.10 m
Bulk velocity W_b	0.45 m/s	0.75 m/s
Kinematic viscosity ν	$3.5 \cdot 10^{-4}$ m ² /s	10^{-6} m ² /s
$Re_b = W_b D / \nu$	192.9	75,000
Closure	—	4 $k-\epsilon$ models

Axisymmetric pipe solved with PHOENICS. Turbulent closures: standard, RNG, realisable and low-Re $k-\epsilon$. The comparison is specific to the implemented grids and wall treatments.

Grid sensitivity, successive grids

$$\epsilon_{i,i+1} = \| \varphi_{int} - \varphi_{i+1} \|_{\infty} / \| \varphi_{i+1} \|_{\infty}$$

Developed flow

$$\epsilon_p(z) = \| W(r,z) - W_{ref}(r) \|_{\infty} / \| W_{ref} \|_{\infty} \leq 10^{-3}$$

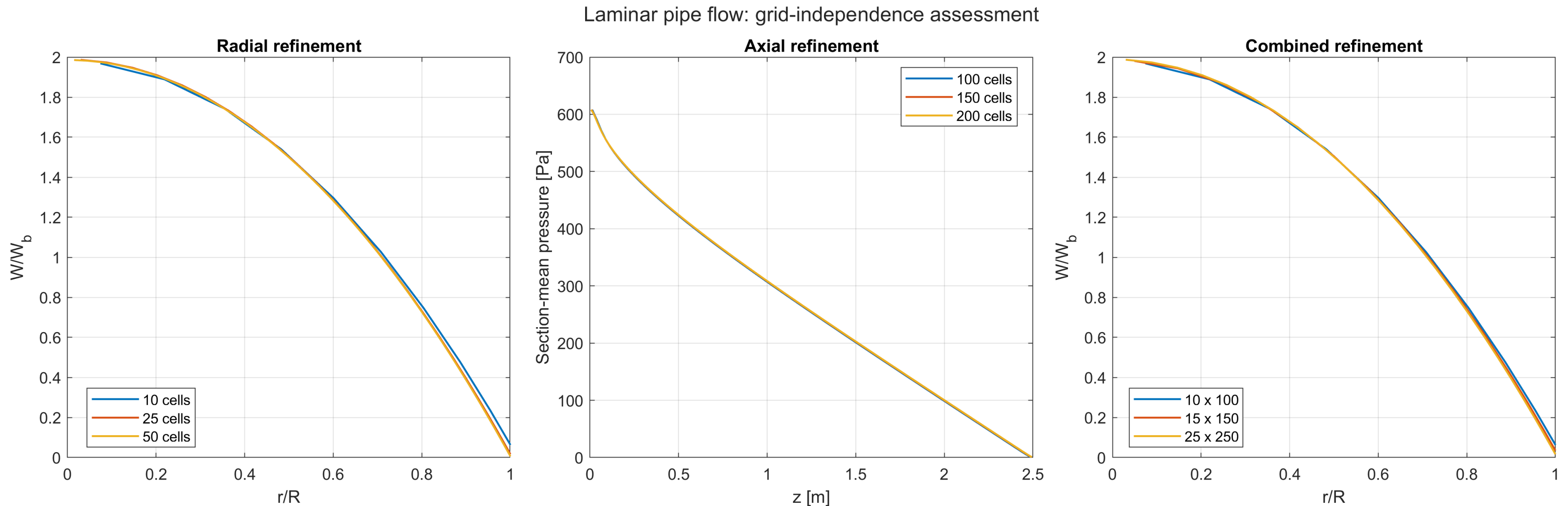
Darcy friction factor

$$f = 8 \tau_w / (\rho W_b^2)$$

Profile validation

Normalised RMSE on experimental radial coordinates

Laminar grid: medium–fine differences below 1%



METHOD

Three grids per path. Radial and combined: velocity profile at $z \approx 0.8L$. Axial: pressure at half radius. Coarse result interpolated onto the finer grid.

OBSERVATION

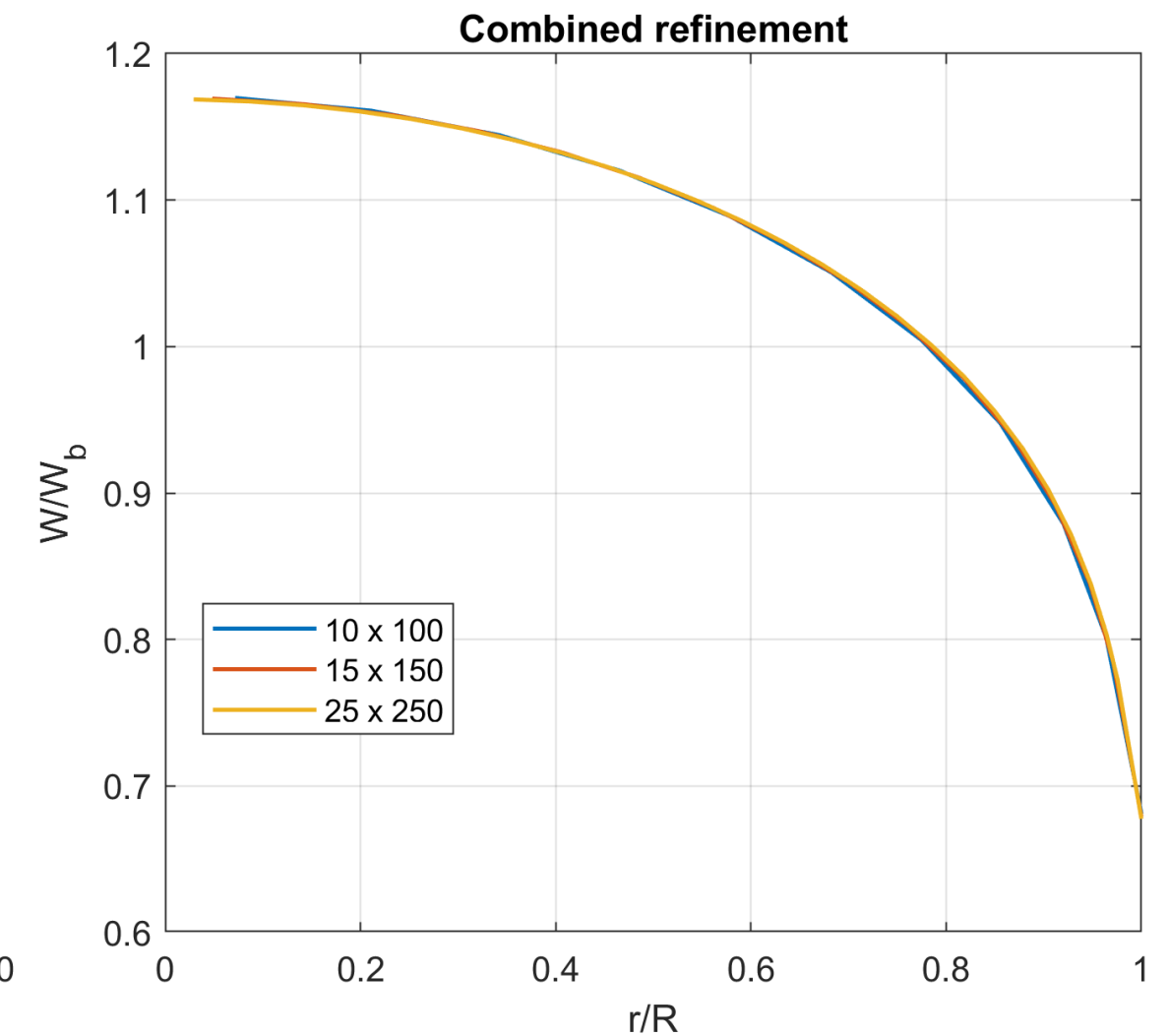
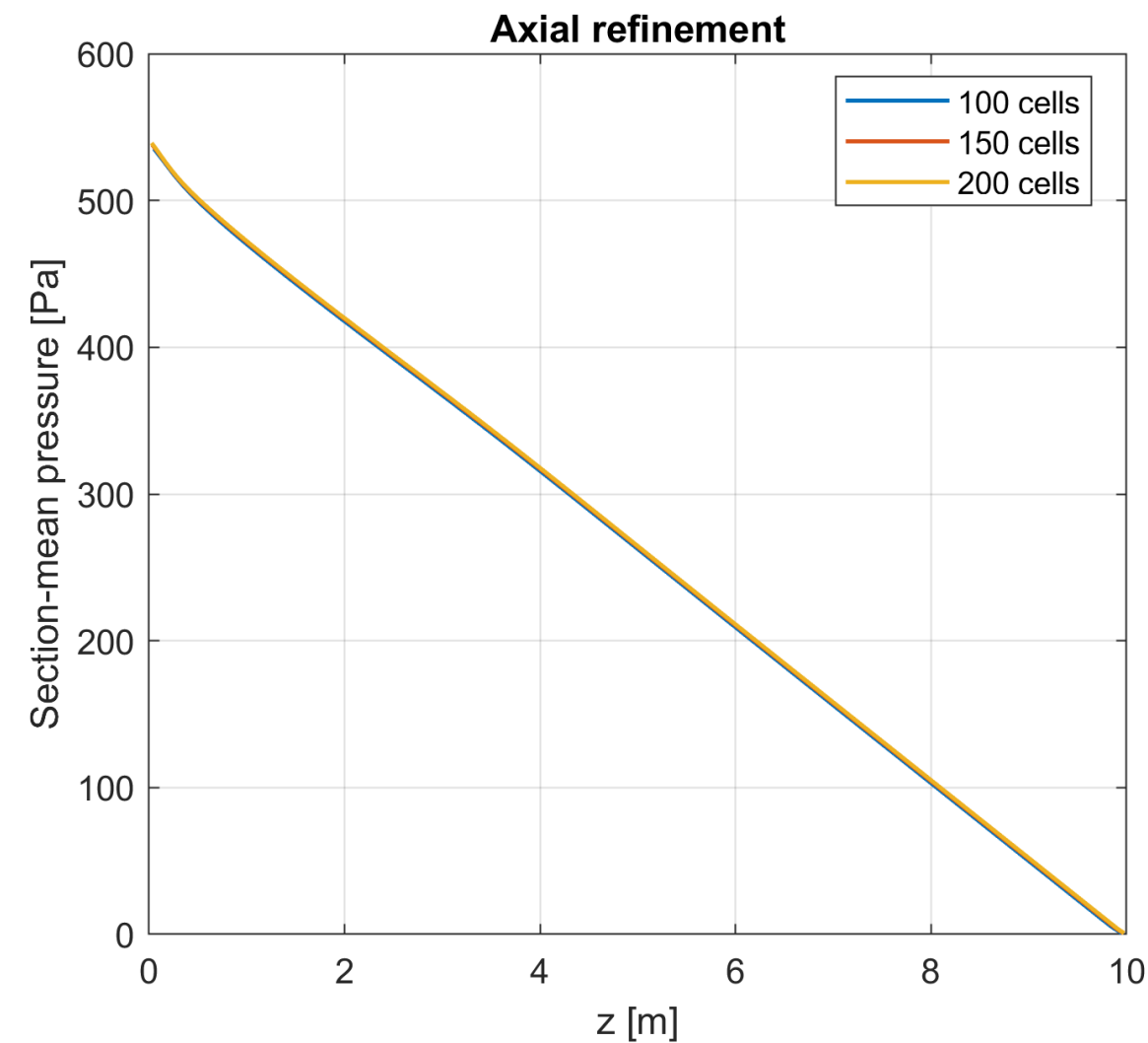
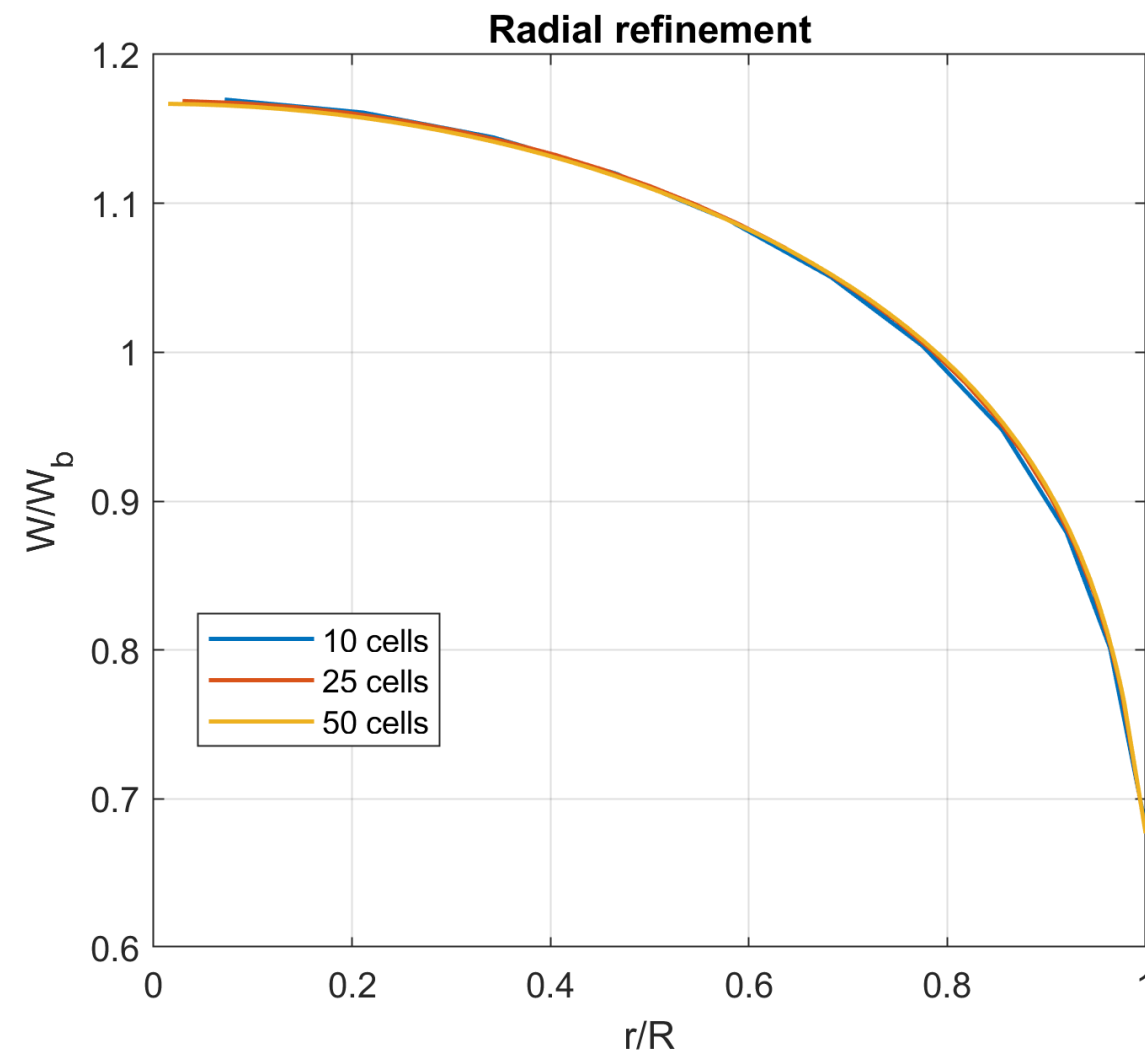
Radial $2.36 \rightarrow 0.52\%$, axial $0.41 \rightarrow 0.23\%$, combined $1.44 \rightarrow 0.93\%$. No change in shape and no spurious oscillations.

IMPLICATION

The finest grid is retained. Pairwise differences, not a formal GCI: the refinement ratio is not uniform across directions.

Turbulent grid: lower sensitivity than the laminar case

Turbulent pipe flow: grid-independence assessment



GRIDS

Radial 10 / 25 / 50 cells, axial 100 / 150 / 200 cells, combined 10×100 → 25×250.

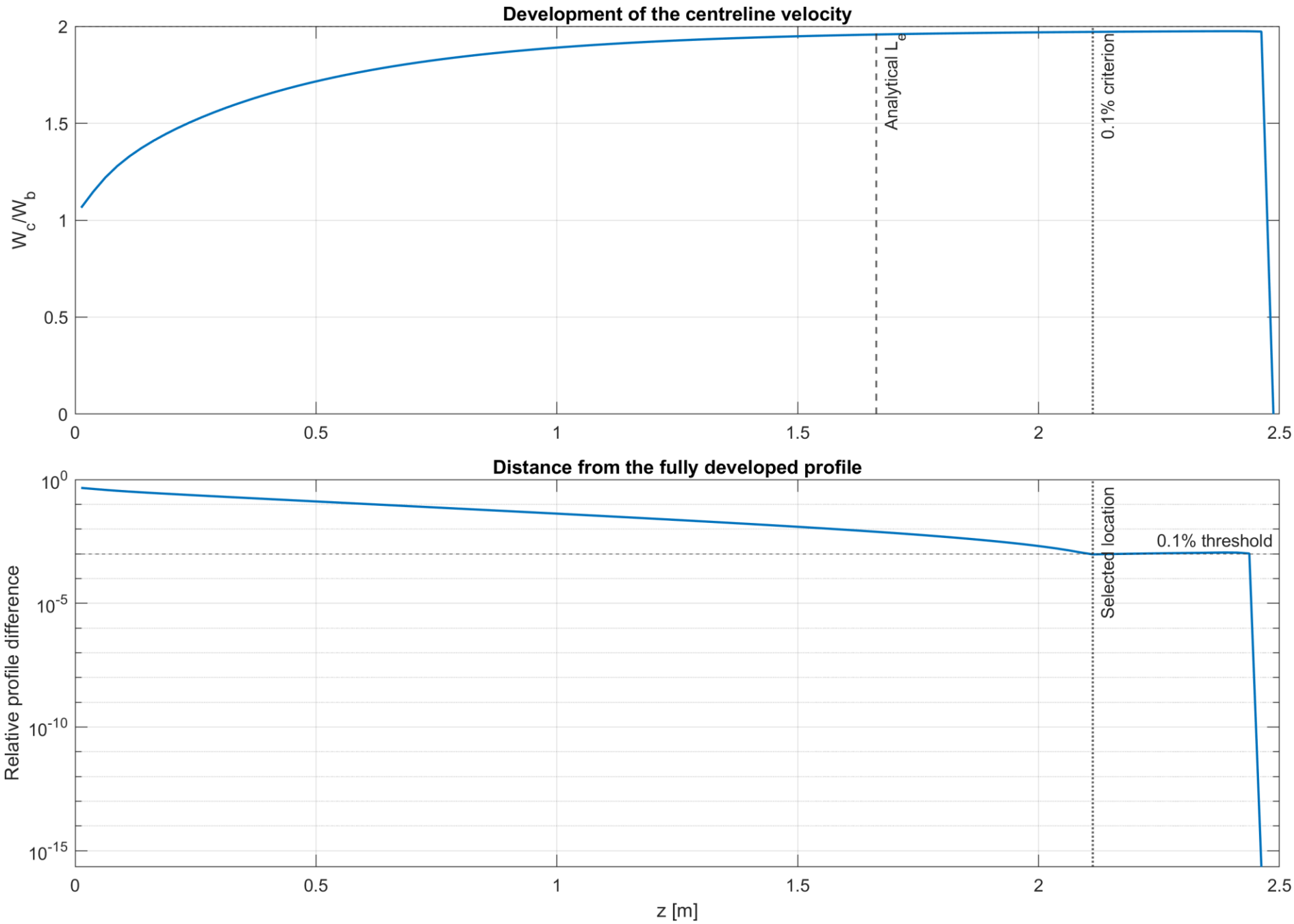
OBSERVATION

Medium–fine differences 0.24% radial, 0.13% axial, 0.37% combined. Residual differences concentrate near the wall.

IMPLICATION

The tested refinements produce small profile changes. Model differences are interpreted for these grids and wall treatments.

Laminar entry length depends on its definition



CRITERION

First axial station where the full profile is within 0.1% of the last interior profile.

Correlation $0.0575 Re D$

1.663 m

Numerical, 0.1% criterion

2.113 m

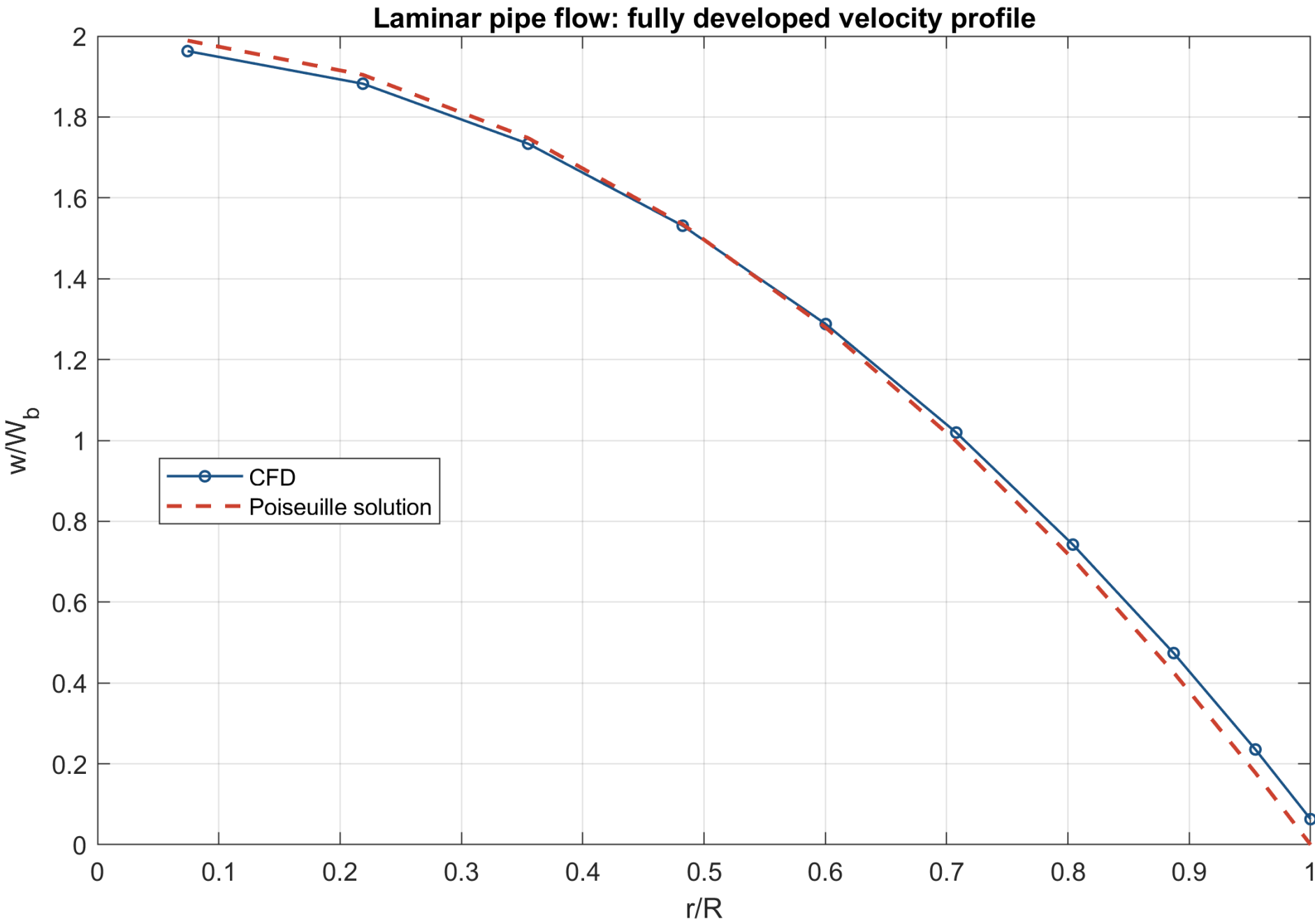
OBSERVATION

Centreline velocity tends to $2W_b$; the profile distance decreases monotonically.

IMPLICATION

The 27% gap reflects the strict full-profile threshold and discrete axial sampling. The asymptotic state is correct.

Developed profile matches Poiseuille within 3.2%



REFERENCE

$$W/W_b = 2[1 - (r/R)^2]$$

Profile L_∞ error

3.18%

Reconstructed W_b

0.4523 m/s · 0.51%

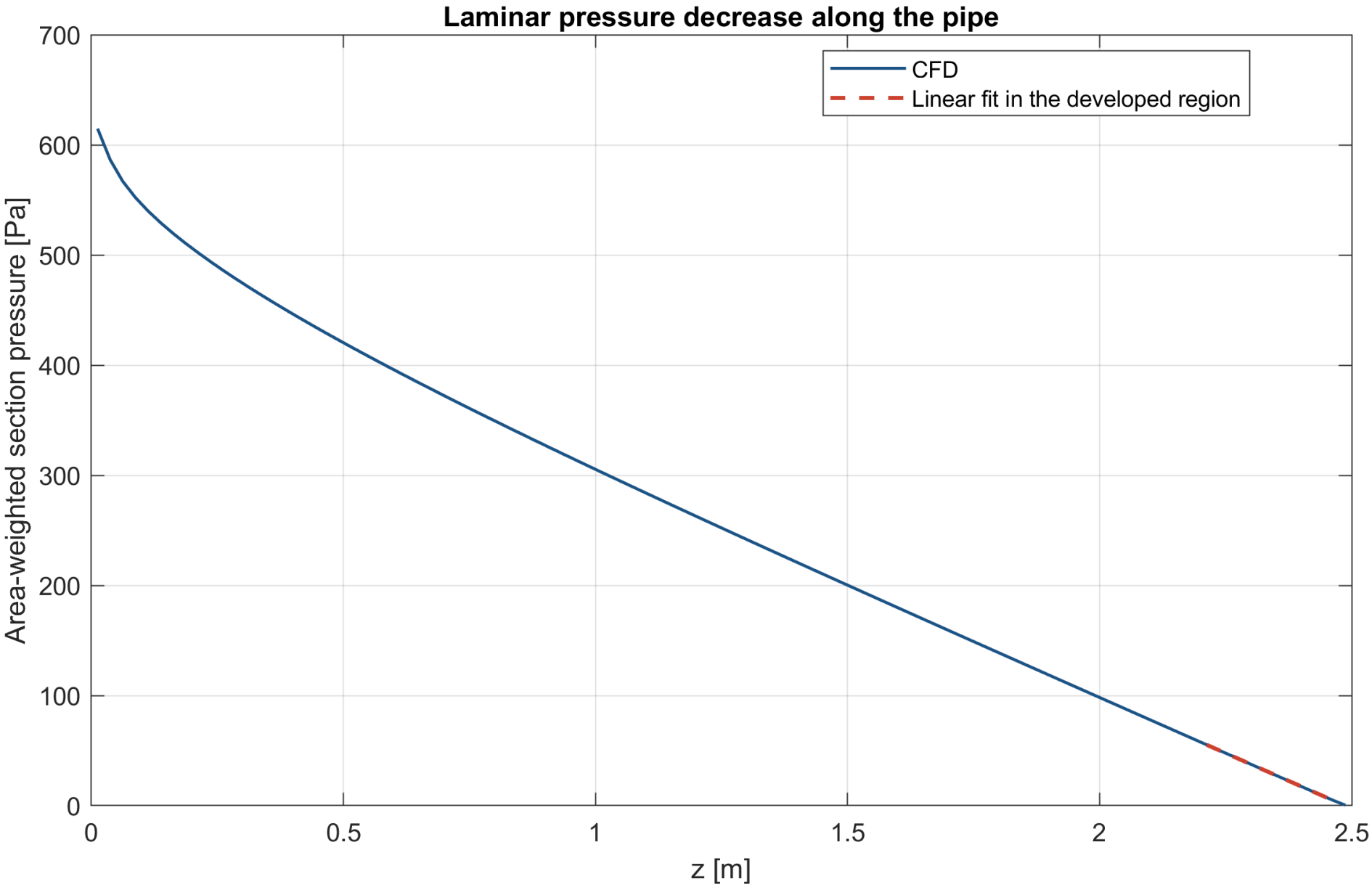
OBSERVATION

The largest differences occur near the wall, where the computed radial profile is steep.

IMPLICATION

The computed profile agrees closely with Poiseuille flow on the tested mesh. The remaining local error has not been isolated.

Pressure gradient matches the analytical value within 1.24%



METHOD

Area-weighted pressure $\bar{p}(z) = (2/R^2) \int_0^R p r dr$; least-squares slope over the developed stations. Reference $-32 \mu W b / D^2$.

Analytical	CFD	Error
-203.84 Pa/m	-201.32 Pa/m	1.24%

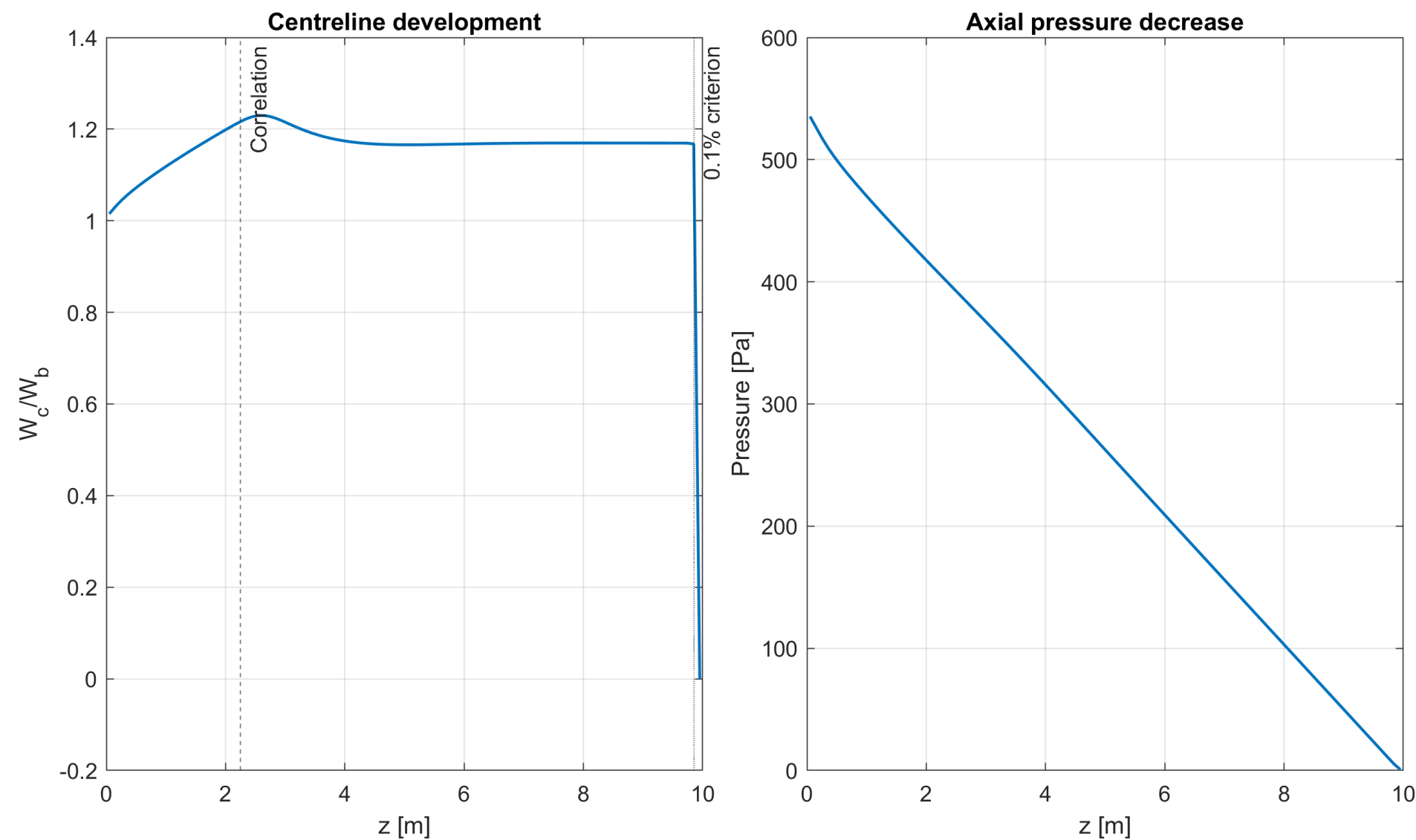
OBSERVATION

Pressure becomes linear once inlet development is complete.

IMPLICATION

The integral momentum balance and the implied wall shear are verified independently of the profile.

Turbulent flow: the centreline converges before the profile



Correlation $1.359 D Re^{1/4}$

2.249 m

Full-profile 0.1% criterion

9.85 m

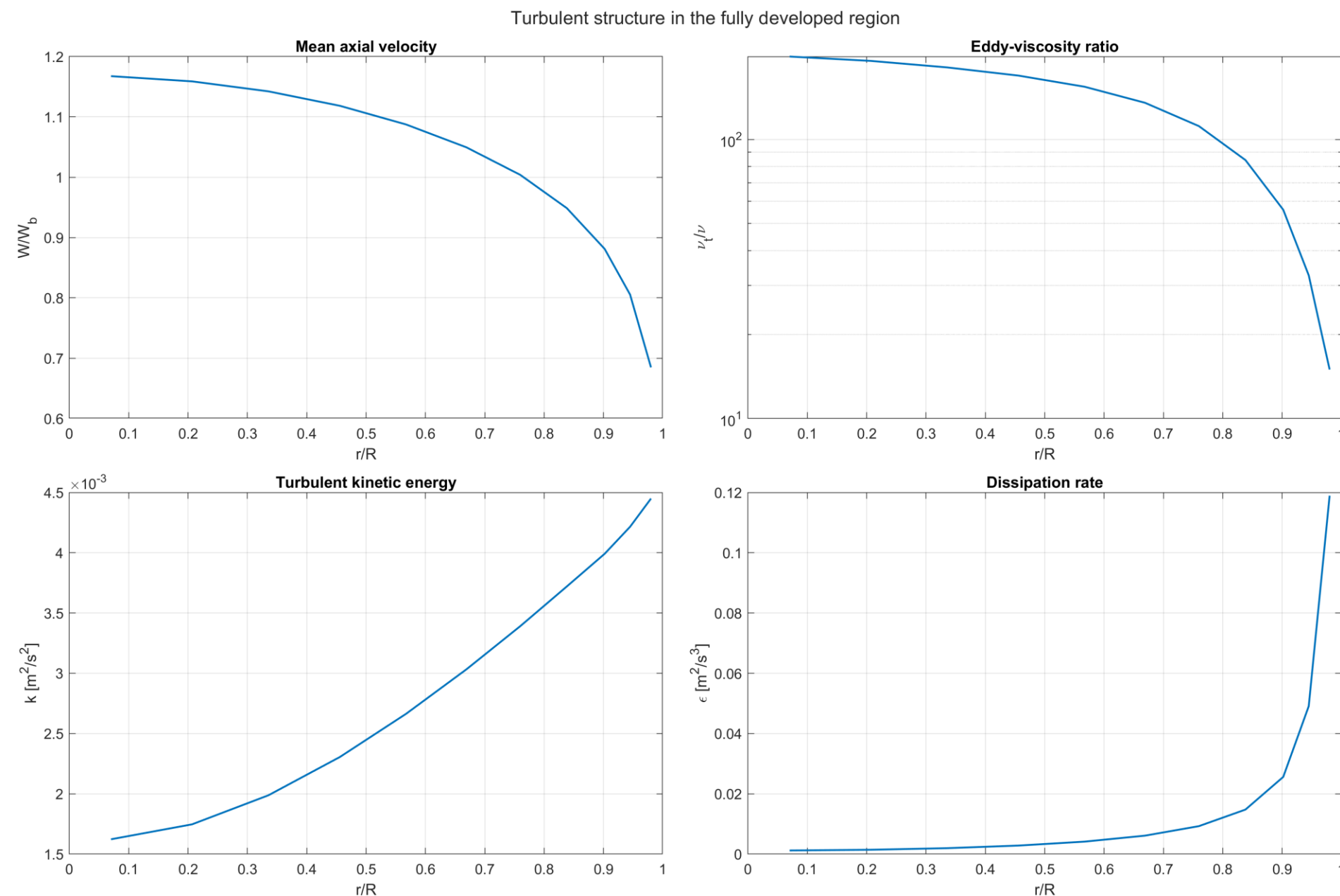
OBSERVATION

Centreline velocity stabilises early; the full radial profile keeps changing weakly downstream. Pressure decreases linearly once developed.

IMPLICATION

A turbulent entry length is not unique: it depends on the observable and on the threshold used to define it.

Developed turbulent field: flat core, active wall region



FIELDS

Mean velocity, eddy-viscosity ratio ν_t/ν , turbulent kinetic energy k and dissipation ϵ in the developed region, standard $k-\epsilon$.

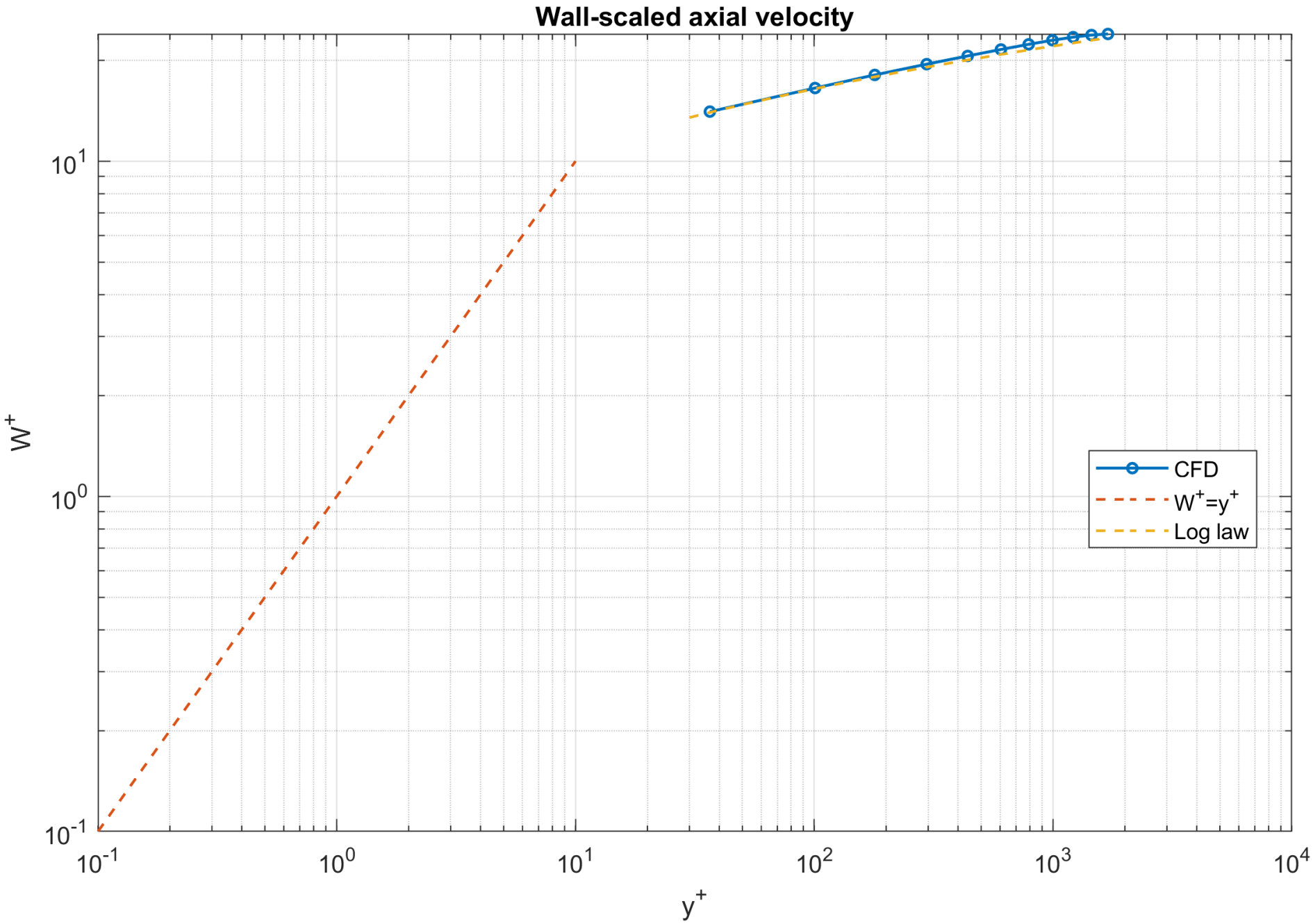
OBSERVATION

The mean profile is flatter than the laminar parabola. k and ϵ increase towards the wall, consistent with shear production. Symmetry about the axis is preserved.

IMPLICATION

The solution is physically consistent before any comparison with measured data.

Inner scaling: the profile crosses the logarithmic range



SCALING

$u_\tau = (\tau_w/\rho)^{1/2}$, $y^+ = y u_\tau/\nu$, $W^+ = W/u_\tau$. Compared with viscous and logarithmic laws.

Friction velocity

0.03654 m/s

Maximum plotted y^+

≈ 1699

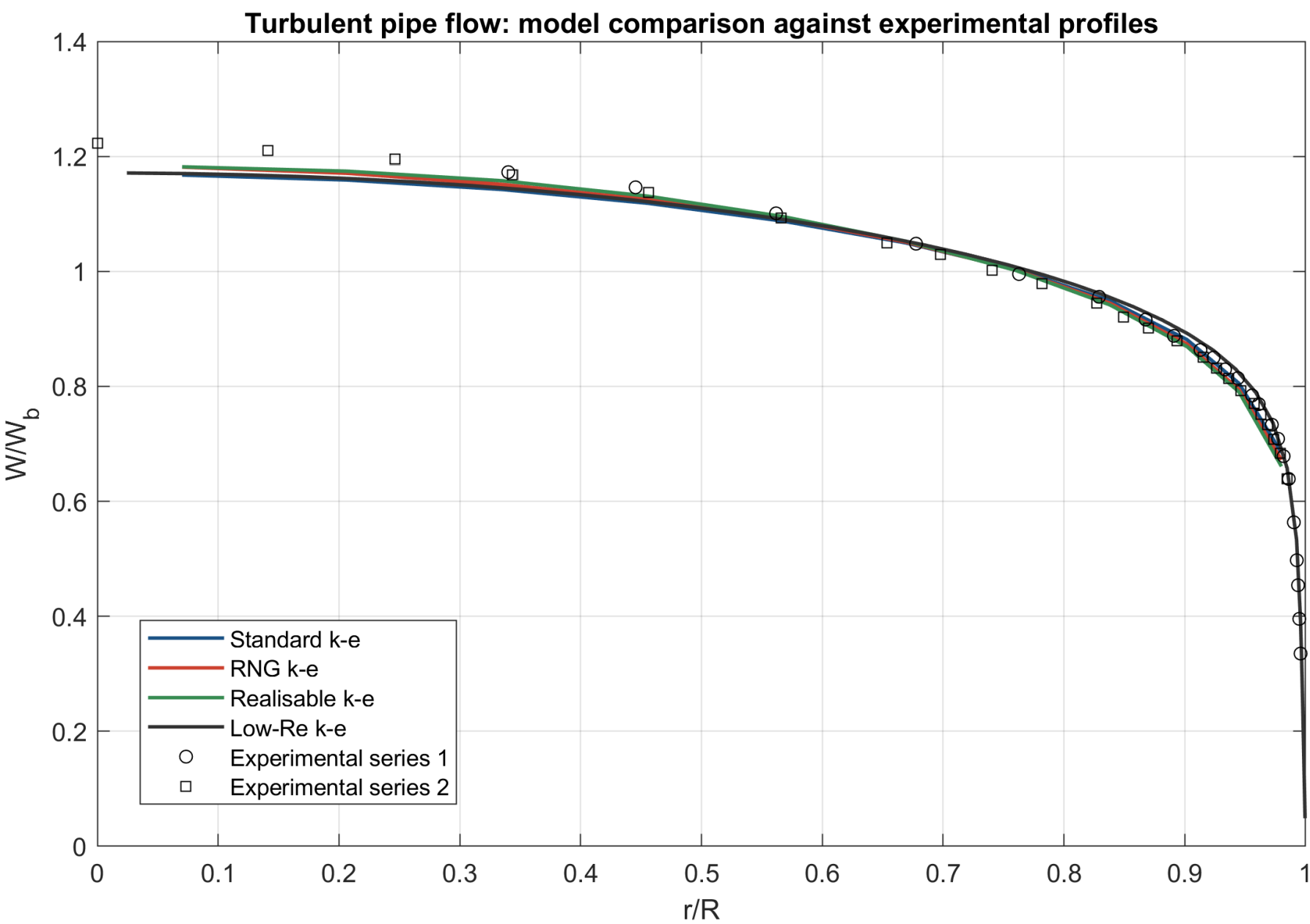
OBSERVATION

The computed profile follows the logarithmic law over the resolved range.

SCOPE

Maximum y^+ is the wall-to-axis distance. The near-wall resolution cannot be reconstructed from the compact exports.

All four closures reproduce the measured profile shape



METHOD

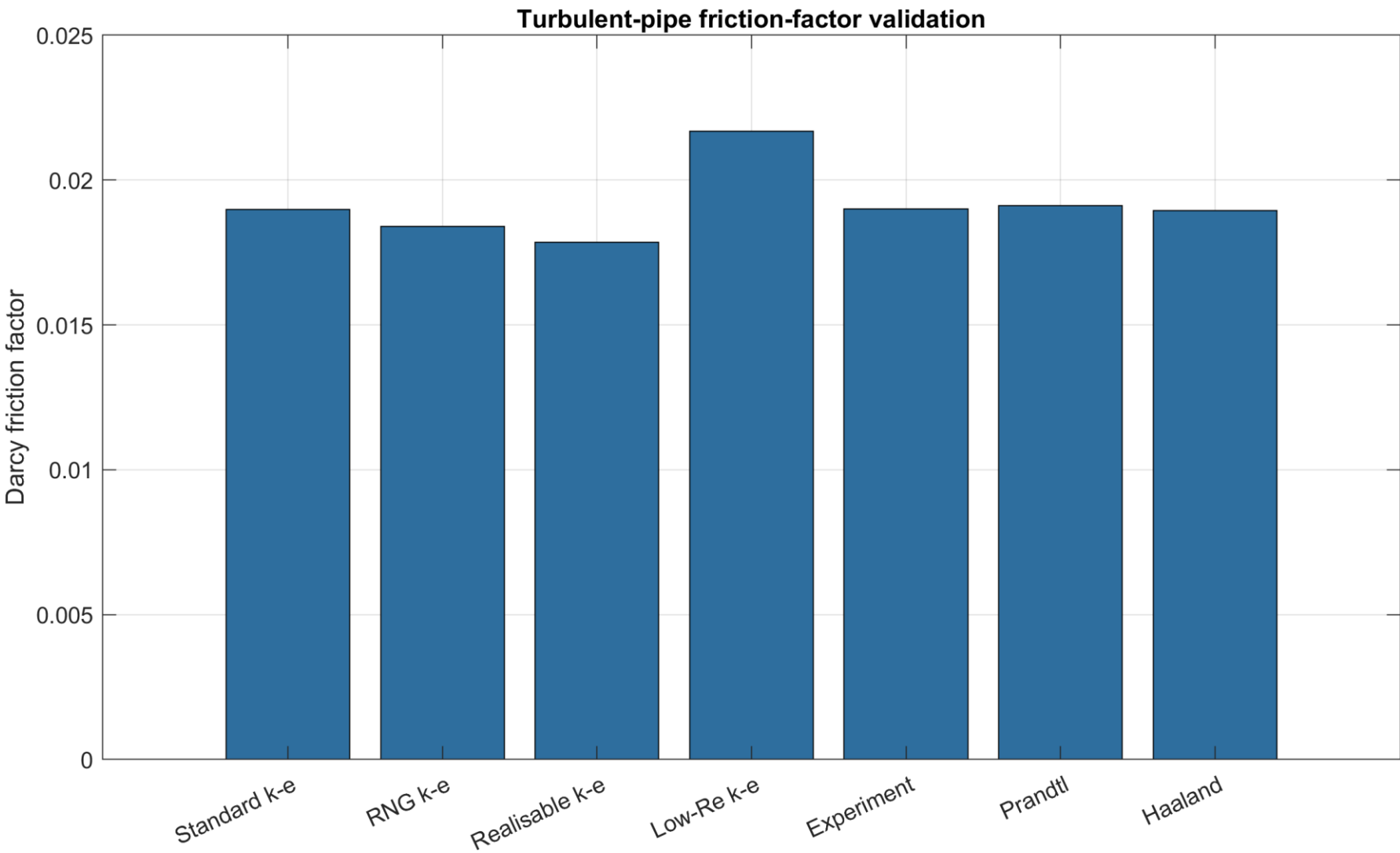
CFD profiles interpolated to the experimental radial coordinates; normalised RMSE against two measured series at $Re_b = 75,000$.

Model	Profile RMSE
Standard $k-\epsilon$	0.0158
RNG $k-\epsilon$	0.0166
Realisable $k-\epsilon$	0.0195
Low-Re $k-\epsilon$	0.0214

IMPLICATION

RMSE spans only 0.016–0.021: profile shape alone weakly discriminates between closures. Wall friction is needed.

Wall friction discriminates between the closures



References: experiment 0.0190 · Prandtl 0.01912 · Haaland 0.01895

Model	f	Error vs exp.
Standard $k-\epsilon$	0.018984	0.082%
RNG $k-\epsilon$	0.018403	3.14%
Realisable $k-\epsilon$	0.017853	6.04%
Low-Re $k-\epsilon$	0.021688	14.15%

INTERPRETATION

The low-Re closure overpredicts wall friction on this grid. Its sensitivity to near-wall resolution requires a dedicated refinement study.

Standard $k-\epsilon$ gives the smallest friction and profile errors for the grids and wall treatments tested.

Cylinder wake experiment

Calibration of the measurement chain, then the wake through its velocity field, its forces and their uncertainty.

18 Experimental setup 19 Load-cell calibration 20 Calibration stability 21 Mean wake
22 Wake recovery 23 Probe spectrum 24 Vorticity phases 25 Cylinder forces 26 Lift spectrum
27 Uncertainty budget

Experimental setup

Circular cylinder, $D = 0.06$ m, in a 0.50 m wide open channel.

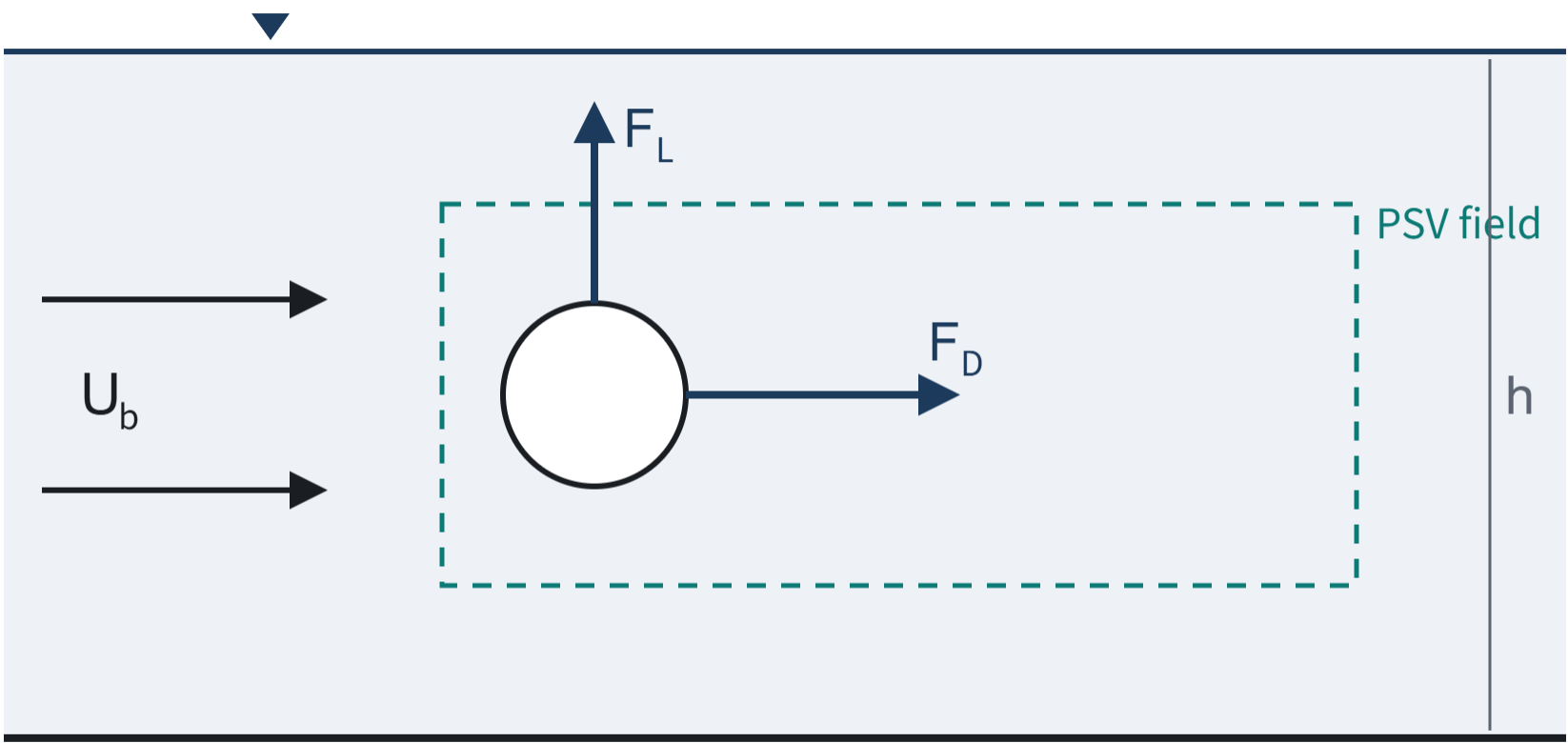
Quantity	PSV	Forces
Flow rate	35 L/s	75 L/s
Water depth	0.42 m	0.45 m
Bulk velocity U_b	0.1667 m/s	0.3333 m/s
Sampling rate	50 Hz	200 Hz
Data	1499 fields, 23×30	span 0.185 m

PSV

Particle-shadow velocimetry: pixel displacements converted at 3040 px/m, local velocity field in the cylinder frame.

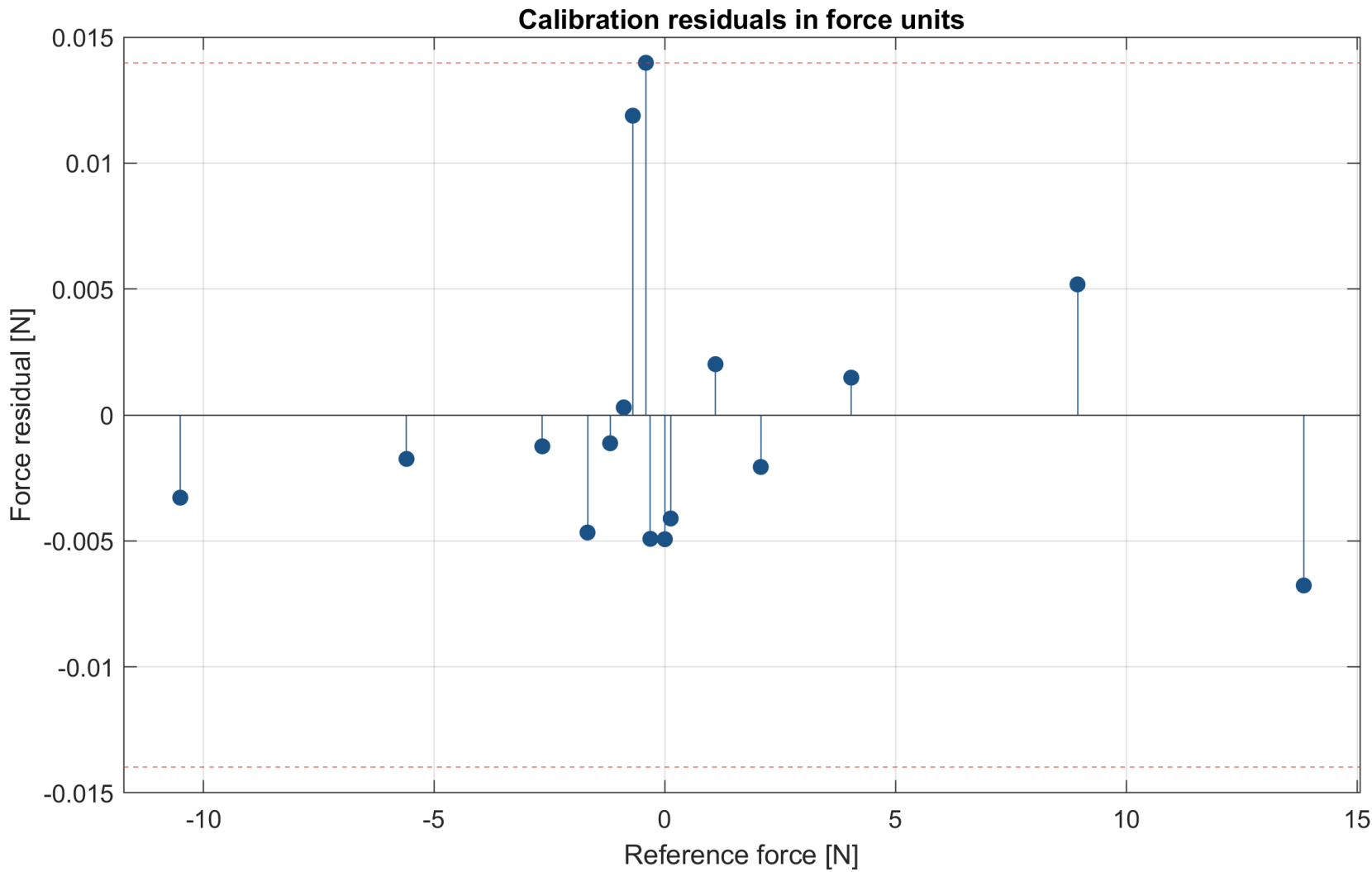
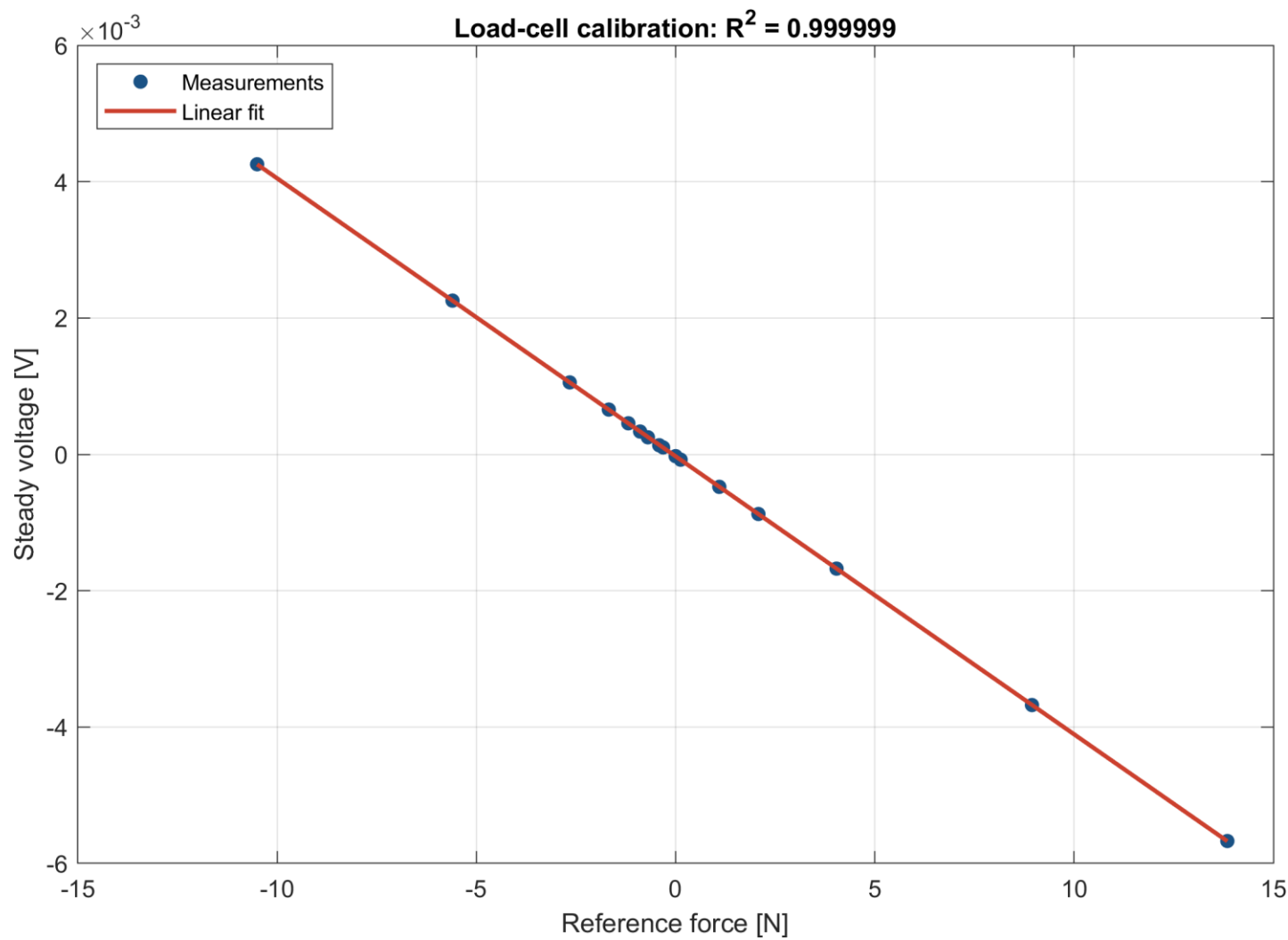
FORCES

Calibrated load cell: span-integrated drag and lift, buoyancy removed with a still-water record.



Side view · free surface, cylinder, measurement plane

Load cell: linear transfer function over -10.5 to 13.8 N



Sensitivity S

$-4.078 \cdot 10^{-4}$ V/N

R^2

0.9999988

Force RMSE

0.00572 N

Max residual

0.01398 N

METHOD

Sixteen static loads at 100 Hz. Fit $V = SF + V_0$ using the last 30% of each record.

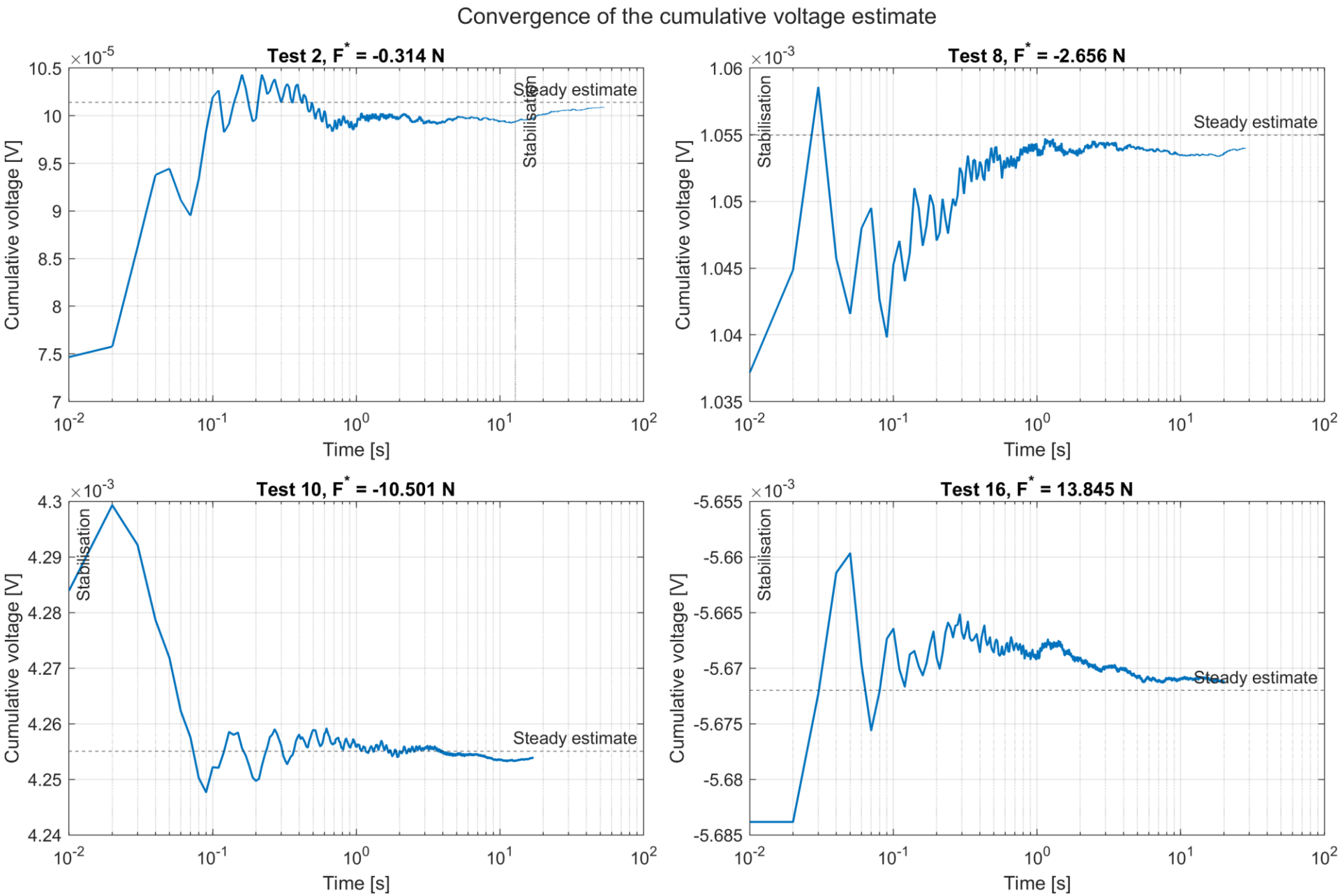
OBSERVATION

No sign-dependent slope or visible curvature. Largest residual near -0.41 N. Leave-one-out refits change sensitivity by less than 0.04%.

IMPLICATION

Small relative to mean drag (0.92 N), but comparable to mean lift (-0.017 N).

Small loads need long records to settle



CRITERION

Cumulative mean within a 2% band of the final level.

Reference load	Settling time
0 N	25.99 s
$ F \leq 0.41$ N	11.9–21.0 s
$ F \geq 0.70$ N	≤ 0.49 s

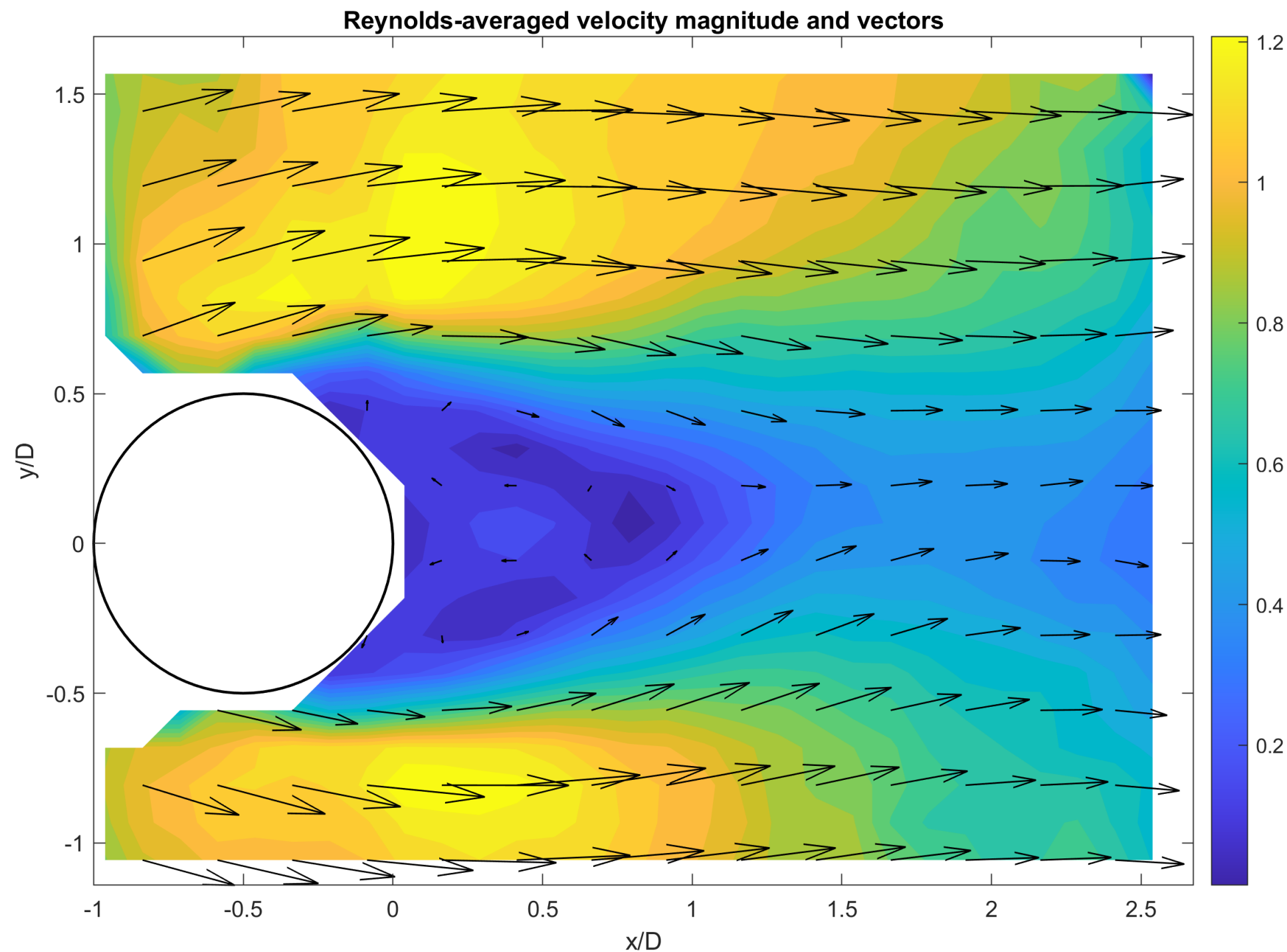
OBSERVATION

High loads settle almost immediately; near zero, drift and noise are comparable with the signal.

IMPLICATION

Justifies the last-30% window and sets the practical limit for interpreting mean lift.

Mean wake: velocity deficit bounded by separated shear layers



Reynolds number

$Re_D = 9980$

Bulk velocity

0.1667 m/s

METHOD

Time average of 1499 PSV fields. Invalid vectors omitted, cylinder interior masked, no spatial interpolation.

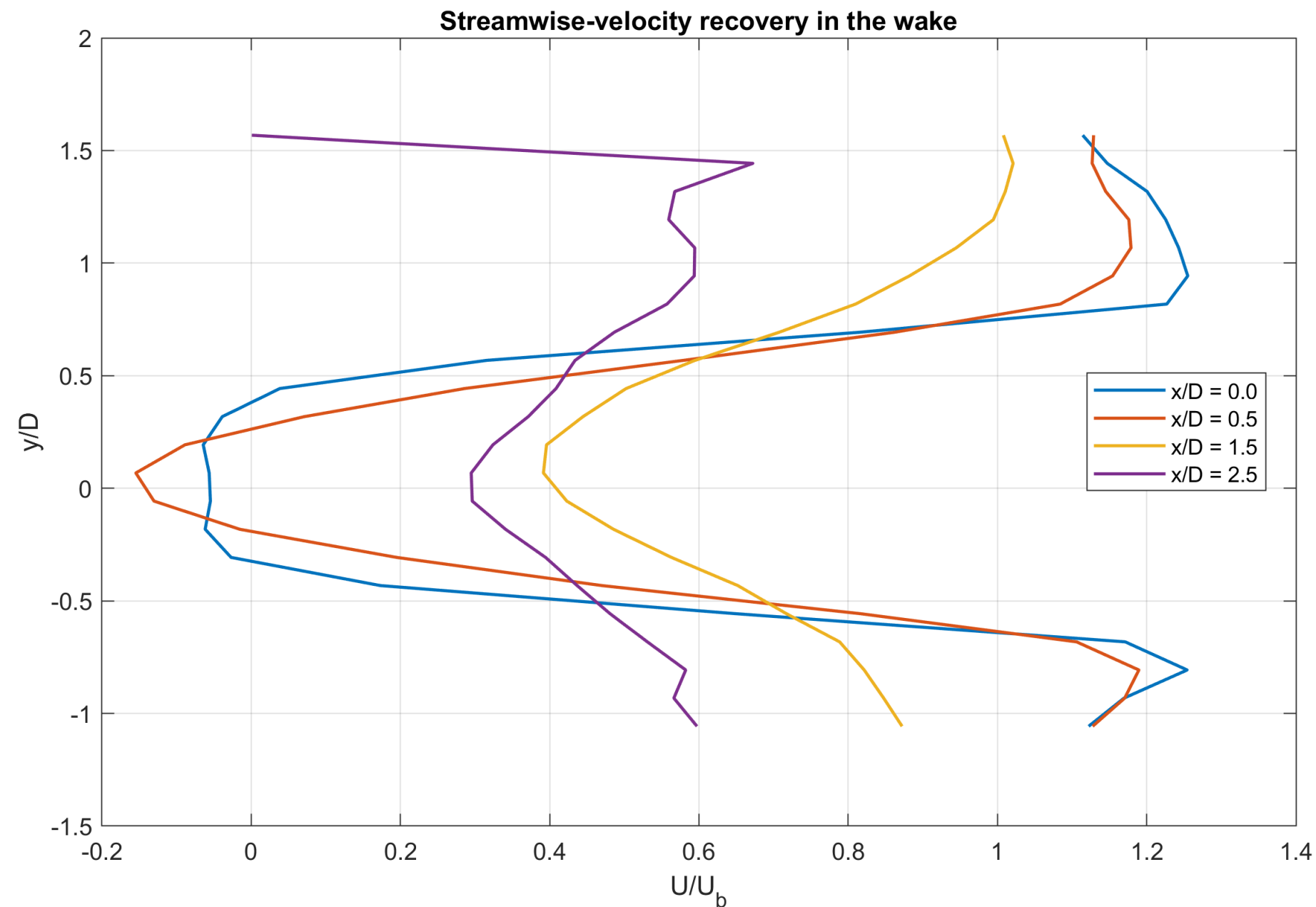
OBSERVATION

Streamwise velocity drops behind the cylinder and a deficit forms around the centreline. Vectors show transverse motion from the shear layers.

SCOPE

Boundary regions with few valid vectors are not interpreted.

Streamwise recovery of the mean velocity



METHOD

Vertical profiles of mean streamwise velocity at four downstream stations.

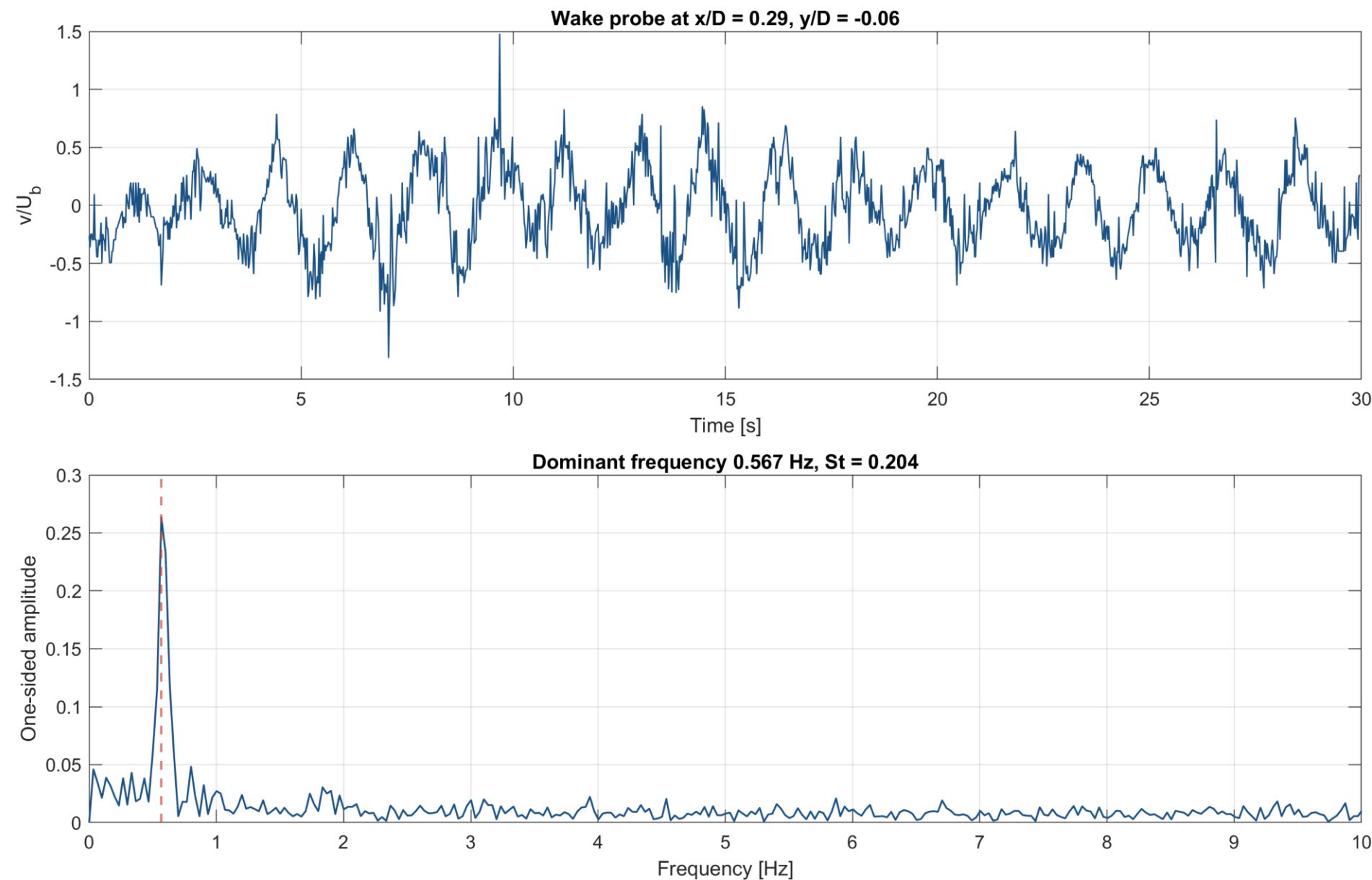
OBSERVATION

The strongest deficit and cross-stream variation occur in the near wake. Profiles smooth out downstream as momentum is redistributed.

SCOPE

The field of view is too short to establish a self-similar far-wake regime.

Velocity spectrum: one dominant frequency, $St = 0.204$



METHOD

Transverse velocity at the valid grid point nearest to $(x/D, y/D) = (0.25, 0)$, i.e. $(0.288, -0.057)$. Gaps interpolated in time; peak searched in 0.2–10 Hz.

Dominant frequency

0.567 Hz

$St = f_s D / U_b$

0.204

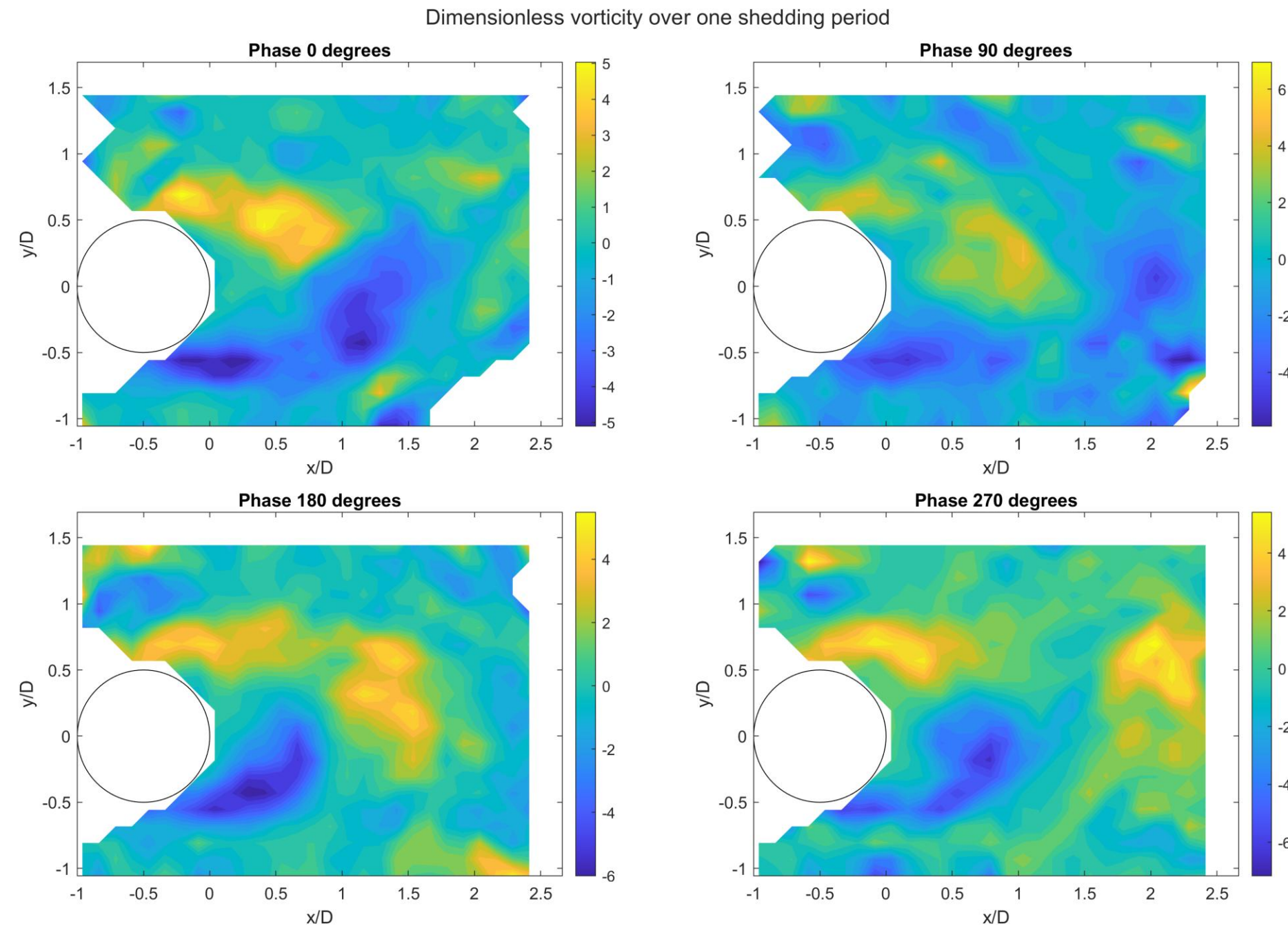
OBSERVATION

A single peak well above the broadband level over a 30 s record at 50 Hz.

IMPLICATION

Consistent with periodic shedding behind a circular cylinder at $ReD \approx 104$.

Vorticity phases: alternating vortices convected downstream



METHOD

Four frames separated by a quarter of the identified period, from a moving-average filtered sequence.

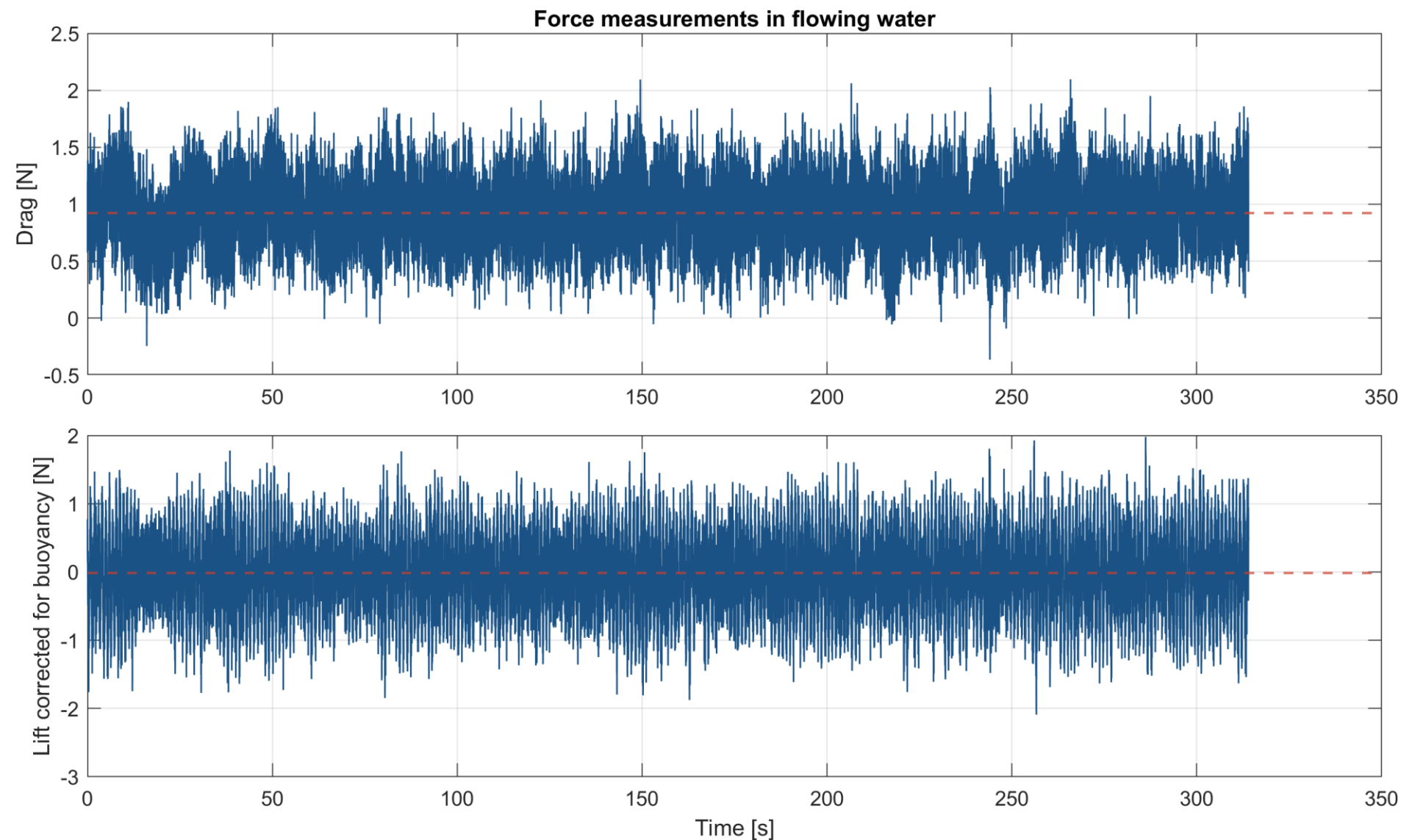
OBSERVATION

Alternating signed structures form behind the cylinder and are convected through the measurement region.

IMPLICATION

Spatial counterpart of the spectral peak.
Topology only: no ensemble phase average, amplitudes not interpreted.

Mean drag coefficient 1.50; mean lift close to zero



METHOD

$C = F / (\frac{1}{2}\rho U_b^2 DL)$, $L = 0.185$ m. Lift corrected for buoyancy with the still-water record.

Mean drag

0.9223 N

Mean lift

-0.0172 N

$\bar{C_D}$

1.4987

$\bar{C_L}$

-0.0280

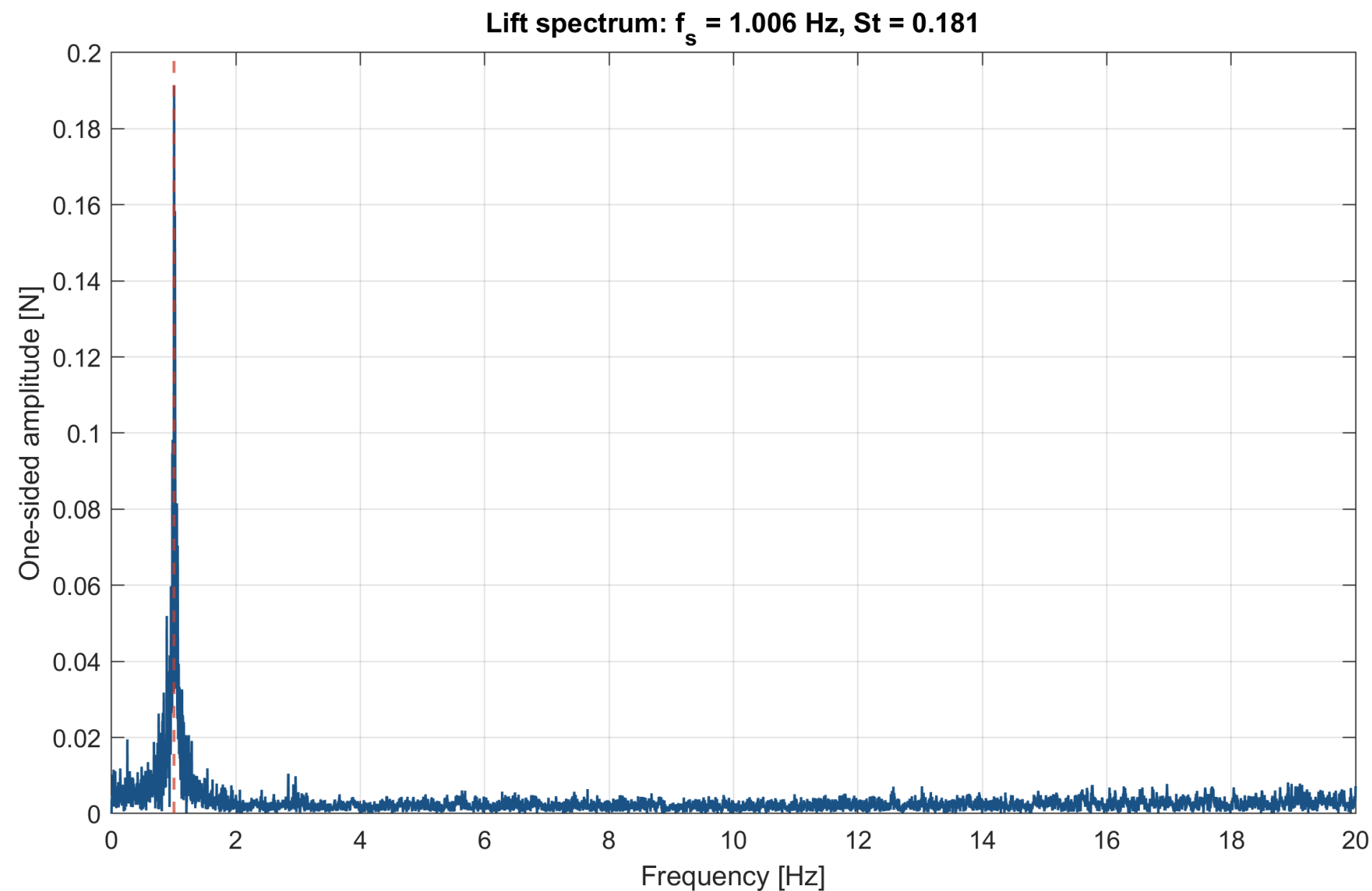
OBSERVATION

Drag fluctuates about a positive mean; lift oscillates about zero.

IMPLICATION

A non-zero mean lift alone is not evidence of physical asymmetry.

Lift spectrum: shedding at $St = 0.181$



METHOD

One-sided amplitude spectrum of the corrected lift after mean removal, 200 Hz.

WHY LIFT

Alternating shedding forces the cylinder cross-stream at f_s ; drag responds mainly at the mean and at even harmonics.

Shedding frequency

1.006 Hz

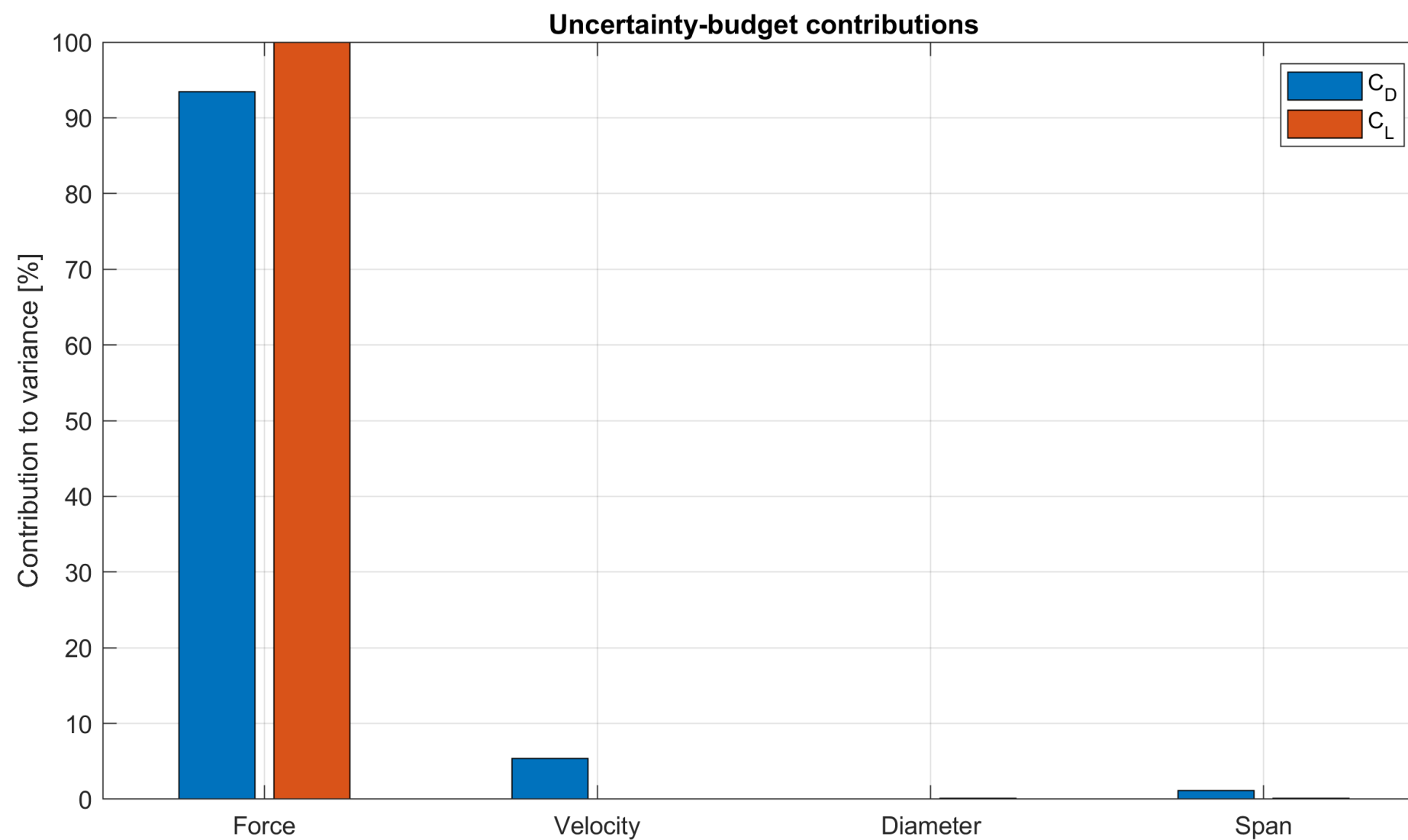
St

0.181

IMPLICATION

The narrow peak indicates a coherent periodic component, measured independently of the velocity field.

Force measurement dominates the uncertainty budget



METHOD

First-order propagation, root sum of squares of independent contributions (assumptions in appendix A2).

$u(C_D)$

0.0756 · 5.0%

$u(C_L)$

0.0574

2 mm level mismatch

$\Delta C_L \approx 0.0229$

OBSERVATION

Force measurement dominates both coefficients. Velocity is next for drag, since $C \propto U_b^{-2}$.

The mean lift is smaller than its uncertainty: it is compatible with zero.

What the evidence supports

Cross-comparison of independent estimates, conclusions and the range over which they hold.

Two Strouhal estimates, one shedding regime

PSV · LOCAL VELOCITY

0.204

$f_s = 0.567$ Hz at $U_b = 0.167$ m/s

LOAD CELL · LIFT

0.181

$f_s = 1.006$ Hz at $U_b = 0.333$ m/s

$|\Delta St| = 0.023$, force estimate 11.3% below PSV

DIFFERENCES BETWEEN THE ACQUISITIONS

- 1 Bulk velocity doubles between the two acquisitions.
- 2 Water depth changes from 0.42 to 0.45 m.
- 3 The estimators differ: local transverse velocity against span-integrated force.

Both estimates indicate the same shedding regime, but the tests were conducted under different flow conditions.

Conclusions

1

LAMINAR CFD · VERIFIED

Final refinement changes the observables by less than 1%. Profile error **3.18%**, pressure-gradient error **1.24%**.

3

MEASUREMENT CHAIN

Linear load cell, $R^2 = 0.9999988$. Near-zero loads settle slowly and bound the interpretation of mean lift.

2

TURBULENT CFD · VALIDATED

Standard $k-\varepsilon$ gives the lowest profile RMSE and $f = 0.018984$, **0.082%** from experiment.

4

CYLINDER WAKE

$\bar{C}_D = 1.499 \pm 0.076$, $\bar{C}_L = -0.028 \pm 0.057$. Coherent shedding at **$St = 0.181-0.204$** .

Each result is stated together with its reference, its error and the conditions under which it holds.

Range of validity and extensions

CONDITIONS OF THE RESULTS

Grid	Sensitivity quantified pairwise, below 1% on the final refinement in both regimes
Closures	Model ranking established for the implemented mesh and wall treatment
Wake field	Mean field and shedding frequency quantitative; vorticity phases describe topology
Forces	Instrument-chain uncertainty propagated; installation effects bounded by a level-mismatch scenario
Strouhal	Two operating points, compared through nondimensional quantities

EXTENSION · OUTCOME

Constant-ratio grid triplets	→ GCI and extrapolated solution
First-cell y^+ control	→ Targeted assessment of the low-Re wall treatment
Conditional phase averaging	→ Quantitative vortex strength
Blockage and free-surface corrections	→ Unbounded-flow force coefficients
Synchronous PSV and force acquisition	→ Direct Strouhal cross-validation at one operating point

The same verification and validation chain extends directly to each case.

Supporting material

Complete grid-sensitivity metrics and the assumptions behind the uncertainty budget.

Complete grid-sensitivity metrics

Regime	Refinement	Coarse–medium	Medium–fine
Laminar	Radial	2.359%	0.521%
	Axial	0.412%	0.225%
	Combined	1.438%	0.925%
Turbulent	Radial	0.842%	0.242%
	Axial	0.260%	0.132%
	Combined	0.625%	0.366%

Relative L^∞ difference between successive grids. Pairwise differences, not an asymptotic GCI.

Uncertainty assumptions and limitations

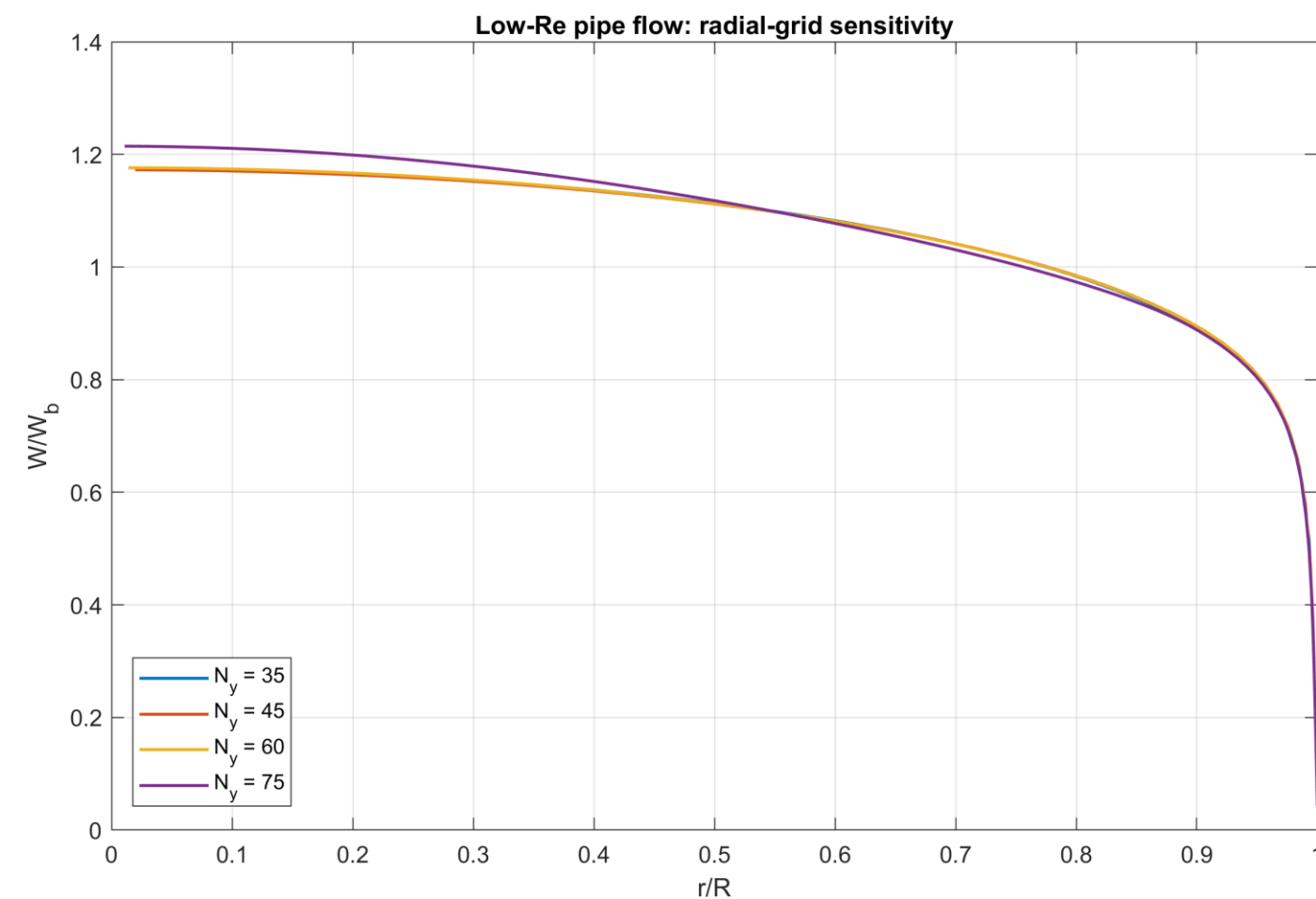
PROPAGATED · FIRST ORDER

Drag force	0.045 N
Each vertical contribution	0.025 N
Depth, channel width	1 mm
Cylinder diameter	0.05 mm
Span	1 mm
Flow rate	0.5% + meter head

NOT PROPAGATED

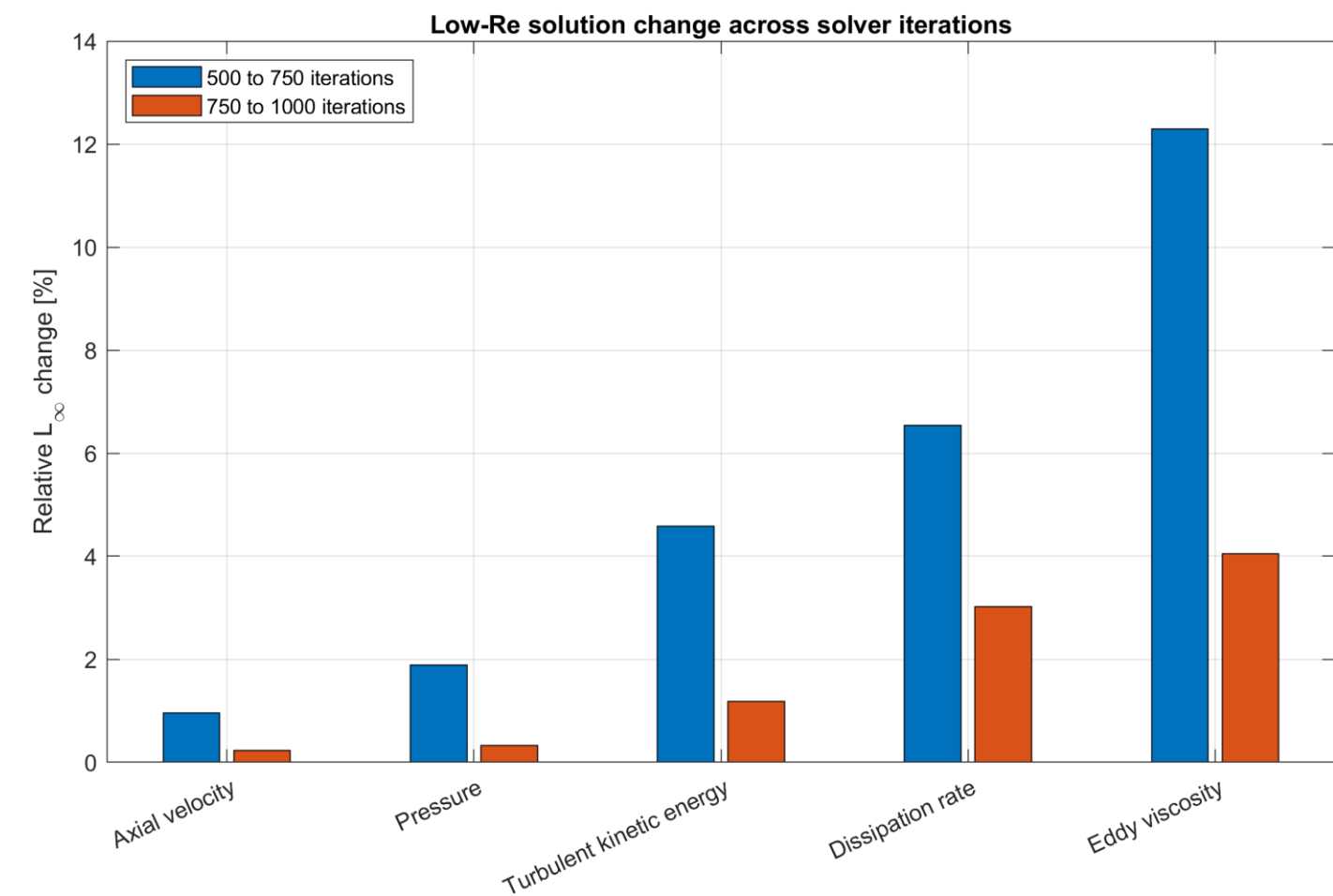
Blockage
Free-surface deformation
End effects
Alignment
Non-simultaneous PSV and force acquisitions

Low-Re closure: resolution remains limiting



RADIAL MESH

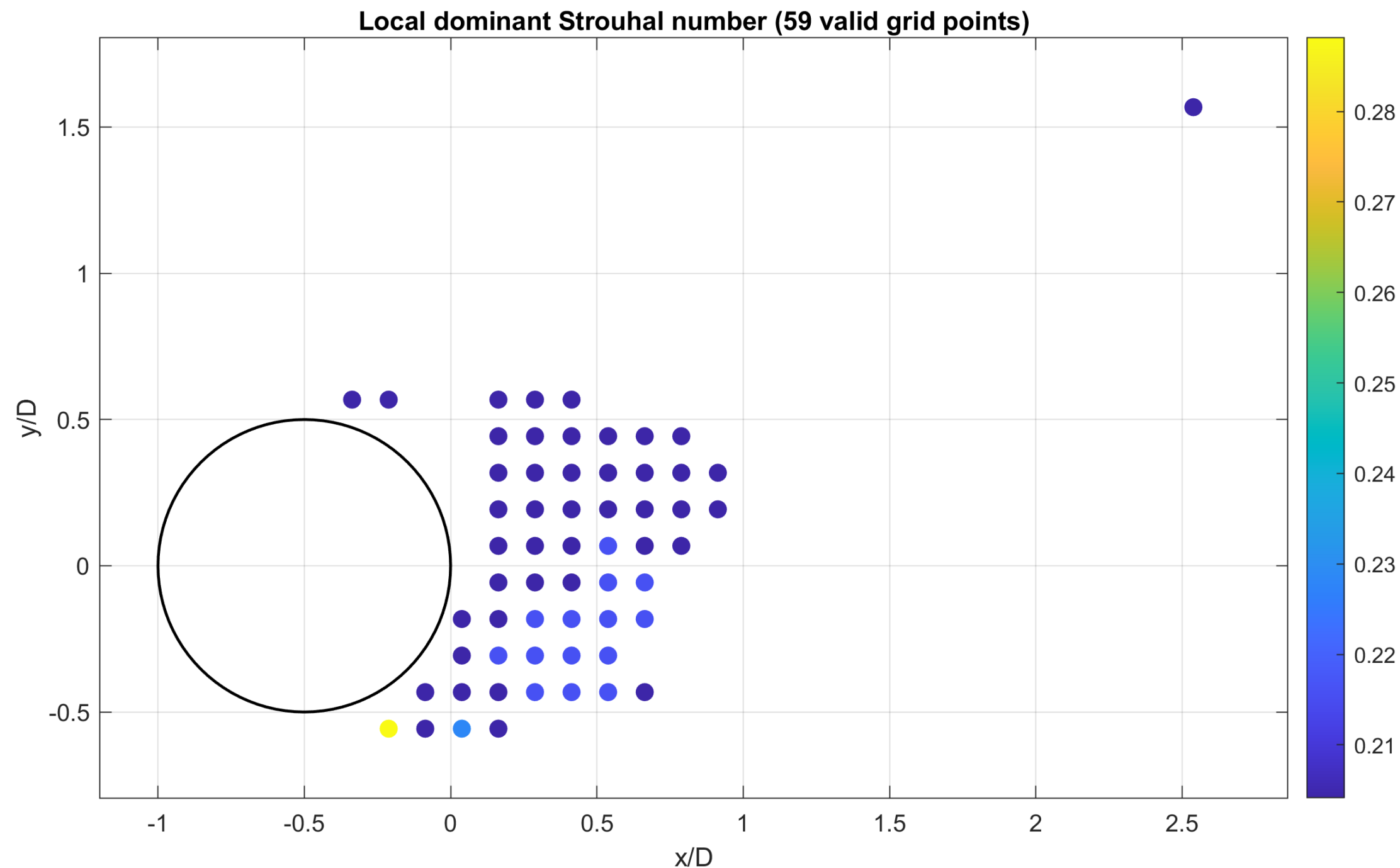
Successive profile changes: 1.10%, 0.55%, then 3.15%. The sequence does not demonstrate monotonic grid convergence.



ITERATION AND SPACING

At 1,000 iterations, eddy viscosity still changes by 4.05%. Spacing-exponent changes are below 0.66%, but do not resolve the mesh trend.

Spatial spectra support the shedding estimate



59 / 690

grid points meet the 95% data-availability criterion.

43 / 59

share the dominant bin at $St = 0.204$.

$\Delta St \approx 0.012$

Finite-record frequency resolution. Isolated higher bins are not interpreted as a spatial gradient.